Software Engineering Teaching Unit

Faculty of Science, University of Kelaniya

A9 Express – Transport Booking Platform

Software Requirement Specification

COURSE UNIT: SENG 31242 – System Design Project

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1.0	2026-05-21	Initial SRS draft	Team
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1. CHAPTER
INTRODUCTION

The A9 Express – Transport Booking Platform is a unified digital transportation management system designed to modernize and simplify transport service bookings in Sri Lanka. The platform integrates private bus ticket reservations, airport transfer services, and customizable tour package bookings into a single online system.

Currently, many transport operators rely on fragmented booking systems, manual coordination methods, and third-party platforms that provide limited flexibility, weak operator branding, and poor customer relationship management. Existing solutions mainly focus on basic seat reservations and lack integrated support for airport transfers and personalized tour services.

The proposed system centralizes transport operations by enabling operators to manage routes, schedules, pricing, bookings, and customer interactions through a dedicated platform. Customers can conveniently reserve transport services, customize Sri Lankan tour packages by selecting tourist destinations, and receive real-time booking updates through a user-friendly interface.

The system aims to improve operational efficiency, strengthen operator brand identity, provide flexible transport solutions, and enhance overall customer experience.

1.1 Purpose

The purpose of this document is to provide a detailed description of the functional and non-functional requirements of the A9 Express – Transport Booking Platform.

This document serves as a reference for developers, stakeholders, project supervisors, and clients to understand the system requirements, functionalities, constraints, and expected behavior before development begins.

The system is intended to provide a centralized transport booking solution that supports:

- Private bus ticket reservations
- Airport pickup and drop booking services
- Customizable Sri Lankan tour packages
- Real-time booking management
- Operator branding and pricing control

Additionally, the system allows users to select tourist attractions within Sri Lanka and automatically generate customized travel packages with calculated pricing based on selected destinations and services.

Intended Audience

- Customers / Passengers
- Company Owners / Managers
- Drivers and Staff
- Tour Operators
- System Administrators
- Software Developers

1.2 Scope

System Name: A9 Express – Transport Booking Platform

The following functionalities shall be provided by the system:

- Manage customer registrations and user accounts
- Enable real-time private bus seat booking
- Manage routes, schedules, and transport availability
- Provide airport pickup and drop booking services
- Allow customers to create customizable Sri Lankan tour packages
- Calculate package pricing based on selected tourist destinations
- Support online booking management and cancellation
- Provide operator branding and pricing customization
- Generate booking confirmations and notifications
- Maintain customer booking history
- Provide analytics and reporting features for operators
- Manage vehicles, drivers, and staff schedules

The following functionalities shall not be included by the system:

- International transport booking services
- Airline ticket reservations
- Hotel reservation management
- Multi-country tour management
- Cryptocurrency payment processing

1.3 Definitions, Acronyms, and Abbreviations

Term	Definition
Admin	System administrator responsible for managing the platform
Customer	User who books transport or tour services
Operator	Company managing transport services
Driver	Staff member responsible for transportation services
Booking	Reservation made by a customer
Tour Package	Customized travel package created using selected destinations
Airport Transfer	Transport service for airport pickup and drop
Real-Time Booking	Instant availability and booking updates
GPS	Global Positioning System used for tracking
SRS	Software Requirement Specification
UI	User Interface
Database	Organized collection of application data
API	Application Programming Interface

1.4 Document Overview

A structured specification of the software requirements for the A9 Express – Transport Booking Platform is provided in this document.

- Chapter 1 presents an introduction to the system, including its purpose, scope, definitions, and document overview.
- Chapter 2 describes the project scope, existing problems, proposed solutions, stakeholders, and target users.
- Chapter 3 defines the functional requirements of the system, including booking management, user management, transport operations, and tour package customization.
- Chapter 4 specifies the non-functional requirements related to performance, security, usability, reliability, and scalability.
- Chapter 5 outlines the business rules, assumptions, and system constraints.
- Chapters 6 to 8 present the system models, including use case diagrams, use case descriptions, and activity diagrams.
- Chapter 9 defines the external interface requirements, including user, software, hardware, and communication interfaces.

- Chapter 10 describes the data requirements, including data entities, relationships, and validation rules.
- Chapter 11 discusses alternative solutions and their limitations.
- Chapter 12 evaluates the technical, operational, economic, and schedule feasibility of the system.
- Chapter 13 presents risk analysis and mitigation strategies associated with the project.
- Chapter 14 lists the references used in preparing this document.

2. CHAPTER
PROJECT SCOPE AND STAKEHOLDERS

2.1 Project Scope

The A9 Express – Transport Booking Platform aims to develop a web-based integrated transport management and booking system that modernizes transportation services in Sri Lanka. The platform consolidates private bus reservations, airport transfer bookings, and customizable tour package management into a single, unified digital solution.

Key goals of the system include:

- Eliminate fragmented and manual booking processes used by transport operators
- Provide customers with a convenient, real-time online booking experience
- Enable operators to manage routes, schedules, pricing, and vehicles through a centralized dashboard
- Allow customers to build personalized Sri Lankan tour packages by selecting destinations, with automated pricing calculation
- Strengthen operator brand identity through a dedicated platform
- Improve operational efficiency in managing drivers, staff, and vehicle schedules
- Provide analytics and reporting tools to support data-driven decision-making by operators

The system is intended to support multiple transport operators, handling a scalable volume of daily bookings across bus routes, airport transfers, and tour services throughout Sri Lanka.

2.2 Stakeholders

The following stakeholders are involved in or affected by the A9 Express – Transport Booking Platform:

Stakeholder	Role
Company Owner	Primary decision maker and platform owner; controls operator branding, pricing, routes, and business operations
System Administrator	Manages platform-wide configurations, user permissions, system health, and technical operations
Passengers	End users who register, search, and book bus tickets, airport transfers, or customized tour packages
Drivers	Staff responsible for executing transport services; managed through the platform for scheduling and assignments
Tour Operators	Manage and configure tour packages, destinations, and associated pricing
Support Staff	Assist in managing day-to-day bookings, customer queries, and operational coordination
Development Team	Responsible for designing, building, testing, and maintaining the platform
Project Supervisors / Clients	Oversee project progress, validate requirements, and ensure deliverables align with business objectives

2.3 Target Users

The A9 Express platform is designed to serve a diverse range of users who require reliable, affordable, and digitally accessible transport services across Sri Lanka. The primary target users are:

Target User	Description
University Students	Students traveling between cities such as Colombo and Jaffna for academic purposes require affordable, reliable, and easy-to-book transport options.
Business Travelers	Professionals and entrepreneurs who frequently travel for meetings, conferences, or business operations and require punctual, comfortable, and bookable transport services.
Regular Jaffna - Colombo Travelers	Individuals who frequently commute or travel along the Colombo–Jaffna route and need a convenient platform to reserve seats in advance and track schedules in real time.
Tourists & Travelers	Local and international visitors who wish to explore Sri Lankan destinations through customizable tour packages offered on the platform.
Airport Travelers	Passengers requiring reliable airport pickup and drop services coordinated through the platform.

Key characteristics shared across all target users:

- Need for advanced seat reservations to avoid last-minute unavailability
- Preference for digital booking over manual or phone-based methods
- Requirement for real-time updates on schedules, availability, and booking confirmations
- Expectation of transparent pricing and flexible cancellation options

2.4 User Personas

Derived from survey of 100+ transport users in Sri Lanka

AK Arjun Krishnamurthy *Daily Commuter Student · Age 21* Daily Traveller Private Bus Website / App **GOALS**	*"Even booked seats will not be available sometimes — the platform needs to actually hold my seat."* **5 / 5**

Fast, reliable seat booking for daily university commute with instant confirmation and real-time seat availability. **PAIN POINTS** • Seats booked online still taken at the counter • Server slowdowns during peak hours • No real-time seat availability updates	**Seat Availability Priority:** High **Likelihood to Use Platform**

NP Nithya Paramasivam *Weekend Family Traveller · Age 28* Monthly Traveller Private Bus Website **GOALS** Book adjacent seats for family members and receive instant confirmation with easy cancellation options. **PAIN POINTS** • Family members assigned separate seats • No online payment refund on cancellation • Lack of seat-gender labelling	*"Sometimes if we book late there are no seats, or our family gets separated."* **5 / 5** **Importance of Simple UI:** Yes **Would Recommend**

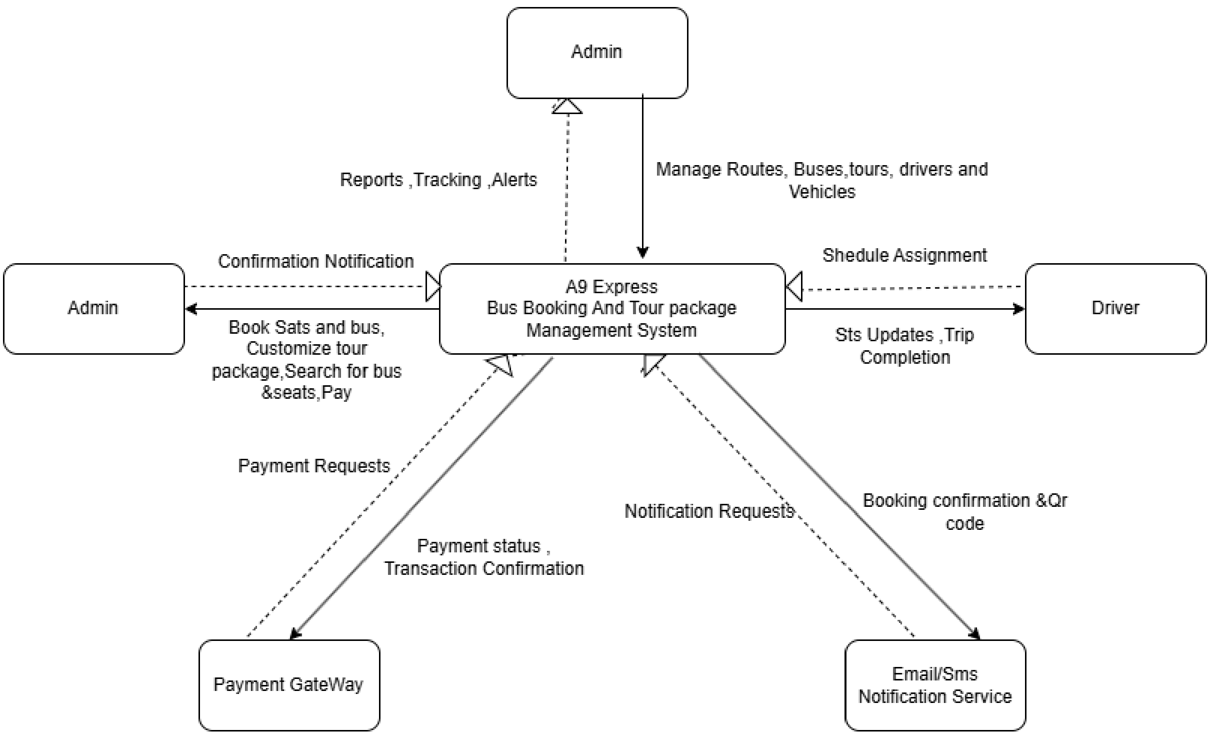
RS Rajan Subramaniam *Working Professional · Age 30* Weekly Traveller Private Bus Mobile App **GOALS** Book work-related transport quickly on mobile with transparent pricing and minimal steps. **PAIN POINTS** • Hidden fees and surge pricing • Too many steps to complete booking	*"Different prices on different apps for the same ride — just want price transparency and efficiency."* **5 / 5** **Real-Time Availability Need:** High **Likelihood to Use Platform**

TW Thenmozhi Venkataraman *Tourist / Leisure Traveller · Age 26* Occasional Traveller Tour Service Website **GOALS** Customise and book tour packages online in one place — including airport transfers — with all pricing upfront. **PAIN POINTS** • No integrated platform for buses, tours and airport transfers • Fake comfort advertising by operators • Cannot change booking after confirmation	*"I'd love a platform combining bus booking, airport transfers, and tours — just like Airbnb but for transport."* **5 / 5** **Would Use Platform:** Yes **Would Recommend**

KM Karthik Murugesan *Budget-Conscious Student · Age 22* Monthly Traveller Government Bus Website **GOALS** Find cheapest transport option with discounts or loyalty rewards, with a straightforward booking process. **PAIN POINTS** • Service charges on online bookings feel excessive • Booking charges sometimes exceed ticket value • No loyalty or discount programmes available	*"Service charge is very high — discounts for regular users would make a big difference."* **Top Motivation:** Discounts **Likelihood to Use Platform:** Medium

SM Savithri Meenakshisundaram *Safety-Conscious Traveller · Age 24* Weekly Traveller Private Bus Both Platforms **GOALS** Book transport securely with verified driver info, seat-gender options, and reliable payment security. **PAIN POINTS** • No gender-designated seat option available • Payment security concerns on third-party platforms • No driver or vehicle quality rating before boarding	*"While booking, show the quality of the driver and bus — that would build real trust."* **5 / 5** **Trust & Reliability Priority:** Yes **Would Recommend**

2.5 Context Diagram



3. Chapter
System Anlysis

3.1 Fact-Finding Techniques

Two formal fact-finding methods were employed to elicit requirements for the A9 Express digital transport platform, as documented below.

3.1.1 Executive Structured Interview

A structured executive interview was conducted with Mr. A. Nirmalan, Owner of A9 Express, between January and February 2026. The session was facilitated by E. Ronald (Interviewer), with questions prepared by A. Mathumithan and documentation handled by V. Arushanth. The interview lasted approximately 45–60 minutes and aimed at mapping the transition from manual workflows to a centralized digital platform.

Interview Guide (Key Questions)

- What are the main problems you currently face in managing transport bookings?
- What services should the platform mainly support?
- Who are the primary users and what access levels should they have?
- What customer information should be collected during registration and booking?
- How should the booking process work from the customer's perspective?
- What payment methods would you like the system to support?
- How should cancellations, refunds, and rescheduling be handled?
- What type of notifications should customers receive after bookings or cancellations?
- How should routes, schedules, pricing, and seat availability be managed?
- How should drivers, vehicles, and staff be managed within the system?
- Do you require real-time seat availability and booking updates?
- What branding or customization features should be available for the business?
- What future features or improvements would you like the platform to support later?

Key Findings from Interview

- Current Operations: A9 Express manages all bookings manually through phone calls and WhatsApp, resulting in frequent delays, booking errors, and double bookings.

- Core Services Required: The platform must support three primary service types — bus ticket booking, airport transfers, and customized tour packages.
- User Access Levels: Four distinct roles were identified: Customers (make bookings), Operators (manage schedules), Drivers (view assigned trips), and Admins (full system control including driver profiles, vehicles, staff, routes, and pricing).
- Booking Flow: Customers require the ability to search routes or services, select seats, confirm bookings, make payments, and receive automated notifications seamlessly.
- Payment Methods: The system must support credit/debit cards, online banking, and mobile payment methods.
- Cancellation & Rescheduling: Customers should be allowed to cancel or reschedule within a policy-defined time period, with automatic refund processing.
- Notifications: Automated booking confirmations, cancellation notices, and trip reminders must be delivered via email or SMS.
- Real-Time Updates: Seat availability and booking status must update in real time across the platform.
- Branding & Customization: The platform should allow admins to customize the company logo, name, contact details, and pricing.
- Reporting: Admin dashboards must include reports on bookings, revenue, vehicle usage, and popular routes.
- Future Enhancements: Planned future features include mobile applications, GPS tracking, multilingual support, and loyalty programs.

3.1.2 Online Customer Survey

An online survey was distributed to potential customers of A9 Express to gather quantitative data on travel habits, booking preferences, platform expectations, and trust factors. A total of 118 responses were collected, predominantly from University Students (83.1%) along with Working Professionals (11.9%), Family Travelers (7.6%), and Tourists (3.4%).

<u>Survey Instrument (Key Question Areas)</u>

- Frequency of private transport use
- Primary reason for travel
- Transport services most frequently used
- Current booking methods
- Difficulties faced when booking transport services

- Dislikes about current transport booking platforms
- Importance of real-time seat availability
- Experience with delays or inconveniences
- Satisfaction with current booking experiences
- Preferred platform type (Website / Mobile App / Both)
- Services preferred for online booking
- Most important features in a booking platform
- Additional services desired
- Concerns about online booking platforms
- Factors encouraging use of an online booking platform
- Interest in a unified integrated booking platform

Key Findings from Survey (118 Responses)

- Travel Frequency: 38.1% of respondents travel monthly, 27.1% rarely, 22% daily, and 21.2% weekly, indicating a broad and recurring need for transport booking solutions.
- Travel Purpose: 73.7% travel for education, making university students the primary target demographic, followed by family visits (27.1%) and work (22.8%).
- Most Used Service: Private buses dominated at 75.4%, confirming bus ticket booking as the highest-priority service for the platform.
- Current Booking Method: 62.7% currently book via website, 37.3% by phone call, 19.5% via mobile app, and 14.4% at a booking office, indicating existing digital readiness among users.
- Booking Difficulties: Respondents reported network issues, availability problems, last-minute changes, and high booking charges as common difficulties. Several noted existing platforms are not user-friendly.
- Platform Dislikes: Common complaints included slow loading times, lack of real-time seat updates, hidden fees, surge pricing, and platforms not meeting international standards.
- Real-Time Availability: 77.1% (scale 4–5) rated real-time seat availability as highly important, making this a critical system requirement.
- Delays and Inconvenience: 41.5% confirmed they have faced delays or inconvenience when booking transport, validating the need for a more reliable digital solution.
- Booking Satisfaction: The majority rated current booking experiences at 3 out of 5 (50.8%), indicating moderate dissatisfaction and significant room for improvement.

- Instant Confirmation: 61% (scale 4–5) rated instant booking confirmation as highly useful, reinforcing the need for automated notifications.
- Preferred Platform: 40.7% prefer both mobile app and website, 30.5% prefer website only, and 28.8% prefer mobile app only — validating the dual-platform (web + app) approach.
- Preferred Services to Book Online: Bus tickets were preferred by 67.8%, followed by all services combined (33.1%), tour packages (14.4%), and airport transfers (5.9%).
- Most Important Features: Easy booking process (75.4%), secure payment (59.3%), real-time availability (50%), and customer support (43.2%) were ranked as most critical.
- Additional Services Desired: Discounts (61.9%), loyalty rewards (50%), tour packages (42.4%), and airport transfers (23.7%) were the most requested additions.
- Key Concerns: Payment security topped concerns at 68.6%, followed by reliability (53.4%), privacy (49.2%), and booking errors (48.3%).
- Encouragement Factors: Easy booking (83.1%), fast confirmation (49.2%), reliable service (38.1%), and discounts (35.6%) were cited as top motivators for platform adoption.
- Unified Platform Preference: 64.2% said YES to a combined platform for bus booking, airport transfers, and tours — directly validating A9 Express's core value proposition.
- Future Usage Likelihood: 71.2% (scale 4–5) indicated they are likely to use such a platform in the future.
- Recommendation Likelihood: 72% indicated they would recommend such a platform to others.

3.2 Existing System Analysis

A9 Express currently operates entirely on manual processes. Bookings are received through phone calls and WhatsApp messages, schedules are tracked informally, and no centralized digital infrastructure exists. The primary pain points of the existing system are analyzed below.

3.2.1 Current Manual Workflow – Pain Points

- Booking Management: All reservations are recorded manually via phone or WhatsApp, leading to frequent double bookings, missed reservations, and human errors with no digital audit trail.
- No Real-Time Seat Visibility: Seat availability is tracked manually, making it impossible to provide customers with live booking status or prevent overbooking during peak periods.
- Payment Handling: There is no integrated payment gateway; customers must make bank transfers or cash payments independently, creating reconciliation difficulties and increasing fraud risk.
- No Automated Notifications: Booking confirmations, reminders, and cancellation notices are sent manually by staff, which is time-consuming and inconsistent.
- Limited Service Scope Visibility: With no platform, customers have no structured way to discover airport transfer or tour package services beyond word-of-mouth or direct calls.
- No Analytics or Reporting: There is no mechanism for tracking revenue, vehicle utilization, popular routes, or booking trends, making business decisions largely intuitive rather than data-driven.
- Driver & Staff Coordination: Trip assignments and schedules are communicated informally, increasing the risk of miscommunication, scheduling conflicts, and operational delays.

- **3.2.2 Competitive Analysis – Feature Comparison Matrix**

A structured competitive analysis was conducted comparing the proposed A9 Express digital platform against the current manual system and comparable transport service platforms to identify feature gaps and market opportunities.

Feature	Manual System (Current)	Competitor A	Competitor B	A9 Express (Proposed)
Online Booking	Yes (basic)	No	No	Yes (full)
Real-Time Seat Availability	No	No	No	Yes
Bus Ticket Booking	Yes	No	No	Yes
Airport Transfer Booking	No	No	No	Yes
Tour Package Booking	No	No	No	Yes
Operator Scheduling Dashboard	No	No	No	Yes
Driver Trip View	No	No	No	Yes
Admin System Management	Limited	No	No	Yes
Online Payment (Card/Bank/Mobile)	No	No	No	Yes
Automated Booking Confirmation (Email/SMS)	No	No	No	Yes
Cancellation & Rescheduling	No	No	No	Yes
Order/Trip Tracking	No	No	No	Yes
Revenue & Analytics Reports	No	No	No	Yes
Business Branding & Customization	No	No	No	Yes

- **3.2.3 Gap Analysis & Justification for the Proposed System**

The competitive analysis and fact-finding sessions collectively establish a clear justification for the development of the A9 Express web and mobile platform. The following gaps were identified:

- No Integrated Digital Platform: No existing solution combines bus ticket booking, airport transfers, and tour packages in a single unified system tailored for the Sri Lankan private transport sector.
- Absence of Real-Time Seat Management: Both the current manual process and comparable alternatives fail to provide live seat availability updates, which 77.1% of survey respondents rated as critical.
- No Role-Based Access Control: The current system lacks structured roles for customers, operators, drivers, and admins, resulting in operational inefficiencies and data security risks.
- No Online Payment Integration: The absence of card, online banking, and mobile payment support limits accessibility and increases transaction risk — a concern noted by 68.6% of respondents.
- Lack of Customer Notifications: Automated booking confirmations and reminders are absent, despite 61% of survey respondents rating instant confirmation as highly important.
- No Business Intelligence: The absence of dashboards, revenue tracking, and route analytics prevents the business from making informed operational decisions.
- No Branding Infrastructure: The current system offers no customer-facing digital identity, limiting A9 Express's ability to build brand trust and compete in the market.

The proposed A9 Express platform addresses all identified gaps by delivering a centralized, role-based, real-time digital solution with integrated payments, automated notifications, analytics, and a unified booking experience across web and mobile channels.

4. CHAPTER
FUNCTIONAL REQUIREMENTS

4.1 User & Access Management

Table 3-1: User & Access Management Functional Requirements

ID	Description	Priority	Source	Status
FR-001	The system shall support multiple user roles: Customer, Operator, Driver, Staff, and Admin, with access controlled using a role-permission matrix.	High	System	Proposed
FR-002	User accounts shall be created, updated, and deactivated by the Admin.	High	Admin	Proposed
FR-003	Role-based permissions shall be configurable and enforceable across all system modules.	High	Admin	Proposed
FR-004	Users shall be authenticated using a valid email and password.	High	System	Proposed
FR-005	Users shall be redirected to role-specific dashboards after successful authentication.	High	System	Proposed
FR-006	User accounts shall be locked after five consecutive failed login attempts.	High	System	Proposed
FR-007	Password reset shall be performed via a time-limited email link.	High	System	Proposed
FR-008	User sessions shall be terminated after 30 minutes of inactivity.	Medium	System	Proposed

4.2 Customer Registration & Profile Management

Table 3-2: Customer Registration Functional Requirements

ID	Description	Priority	Source	Status
FR-009	Customers shall be able to register by providing their name, email, phone number, and password.	High	Customer	Proposed
FR-010	Customer profiles shall be updatable including contact details and preferences.	High	Customer	Proposed
FR-011	Customer booking history shall be viewable from the customer profile.	High	Customer	Proposed

4.3 Bus Ticket Booking

Table 3-3: Bus Ticket Booking Functional Requirements

ID	Description	Priority	Source	Status
FR-012	Customers shall be able to search for available buses by selecting origin, destination, and travel date.	High	Customer	Proposed
FR-013	The system shall display available seats in real-time for each bus route.	High	System	Proposed
FR-014	Customers shall be able to select one or more seats and confirm a booking.	High	Customer	Proposed
FR-015	The system shall prevent double-booking of the same seat.	High	System	Proposed
FR-016	Booking confirmation shall be sent to the customer via email or SMS upon successful reservation.	High	System	Proposed
FR-017	Customers shall be able to cancel a bus booking within the permitted cancellation window.	High	Customer	Proposed
FR-018	Booking status shall be maintained as Confirmed, Cancelled, or Completed.	High	System	Proposed

ID	Description	Priority	Source	Status
FR-019	An incomplete bus booking shall be automatically cancelled by the system if payment or confirmation is not completed within 15 minutes of initiation.	High	System	Proposed

4.4 Route & Schedule Management

Table 3-4: Route & Schedule Management Functional Requirements

ID	Description	Priority	Source	Status
FR-020	Admin shall be able to create, update, and deactivate bus routes including origin, destination, and stops.	High	Admin	Proposed
FR-021	Admin shall be able to configure schedules including departure times, days of operation, and seat capacity.	High	Admin	Proposed
FR-022	The system shall automatically update seat availability when bookings are made or cancelled.	High	System	Proposed
FR-023	Admin shall be able to set and adjust ticket pricing per route.	High	Admin	Proposed

4.5 Airport Transfer Booking

Table 3-5: Airport Transfer Booking Functional Requirements

ID	Description	Priority	Source	Status
FR-024	Customers shall be able to book airport pickup and drop-off services by specifying date, time, location, and flight details.	High	Customer	Proposed
FR-025	The system shall display available vehicles and estimated transfer pricing for airport transfers.	High	System	Proposed
FR-026	Airport transfer booking confirmation shall be sent to the customer upon successful reservation.	High	System	Proposed

ID	Description	Priority	Source	Status
FR-027	Customers shall be able to cancel an airport transfer booking.	High	Customer	Proposed
FR-028	Admin shall be able to assign a driver and vehicle to confirmed airport transfer bookings.	High	Admin	Proposed

4.6 Tour Package Customization

Table 3-6: Tour Package Functional Requirements

ID	Description	Priority	Source	Status
FR-029	Customers shall be able to create customized Sri Lankan tour packages by selecting from a list of available tourist destinations.	High	Customer	Proposed
FR-030	The system shall automatically calculate tour package pricing based on selected destinations, vehicle type, and duration.	High	System	Proposed
FR-031	Customers shall be able to save, review, and confirm their customized tour packages before booking.	High	Customer	Proposed
FR-032	Tour package booking confirmation shall be generated and sent to the customer upon successful reservation.	High	System	Proposed
FR-033	Admin shall be able to manage the list of tourist destinations and associated pricing.	High	Admin	Proposed
FR-034	Customers shall be able to view and cancel their tour package bookings.	High	Customer	Proposed

4.7 Vehicle, Driver & Staff Management

Table 3-7: Vehicle, Driver & Staff Functional Requirements

ID	Description	Priority	Source	Status
FR-035	Admin shall be able to add, update, and deactivate vehicles including type, capacity, and registration details.	High	Admin	Proposed
FR-036	Admin shall be able to manage driver profiles including personal details, license information, and status.	High	Admin	Proposed
FR-037	Admin shall be able to assign drivers to specific routes, airport transfers, or tour bookings.	High	Admin	Proposed
FR-038	Staff schedules shall be manageable by the Operator.	Medium	Admin	Proposed
FR-039	Vehicle availability shall be updated automatically when assigned to a booking.	High	System	Proposed

4.8 Operator Branding & Pricing Control

Table 3-8: Operator Branding & Pricing Functional Requirements

ID	Description	Priority	Source	Status
FR-040	Admin shall be able to configure their brand identity including logo, company name, and contact details on the platform.	Medium	Admin	Proposed
FR-041	Admin shall be able to set and update pricing for bus routes, airport transfers, and tour packages independently.	High	Admin	Proposed
FR-042	Admin-specific pricing and branding shall be visible to customers during the booking process.	High	System	Proposed

4.9 Booking Management & Notifications

Table 3-9: Booking Management Functional Requirements

ID	Description	Priority	Source	Status
FR-043	Customers shall be able to view all their current and past bookings from their dashboard.	High	Customer	Proposed
FR-044	The system shall send real-time booking confirmation and cancellation notifications to customers via email or SMS.	High	System	Proposed
FR-045	Admin shall receive notifications when a new booking is made or cancelled.	High	System	Proposed
FR-046	Booking details shall include service type, date, time, passenger details, and payment status.	High	System	Proposed

4.10 Reporting & Analytics

Table 3-10: Reporting & Analytics Functional Requirements

ID	Description	Priority	Source	Status
FR-047	Admin shall be able to generate reports on bookings, revenue, and vehicle utilization.	High	Admin	Proposed
FR-048	The system shall provide an analytics dashboard showing booking trends, popular routes, and revenue summaries.	Medium	Admin	Proposed
FR-049	System-wide reports and audit logs shall be accessible to the Admin.	High	Admin	Proposed

5. CHAPTER
NON-FUNCTIONAL REQUIREMENTS

5.1 PERFORMANCE

Performance requirements define the speed, throughput, and responsiveness standards the A9 Express platform must meet. Given that the target user base includes working professionals (30%), university students (25%), and families (20%) who rely on fast and dependable booking, performance is a critical quality attribute.

ID	Requirement	Measurable Acceptance Criterion	Priority	Source	Status
NFR-P01	Homepage and search results performance	Homepage and search results load within 2 seconds on a standard 4G/broadband connection.	**High**	Client (A9 Express)	*Proposed*
NFR-P02	Seat selection and booking confirmation speed	Pages respond within 3 seconds when 500 concurrent users are active.	**High**	Passenger	*Proposed*
NFR-P03	High concurrency support	System sustains 500 concurrent users with no more than 5% error rate during load tests.	**High**	Admin	*Proposed*
NFR-P04	Airport hire and tour search speed	Search results returned within 3 seconds for any query.	**Medium**	System Design Team	*Proposed*
NFR-P05	Payment processing performance	Transactions complete within 5 seconds under normal load; users notified of failure within 10 seconds.	**High**	Dev Team	*Proposed*

NFR-P06	Mobile app data efficiency	Active use consumes no more than 50 MB of mobile data per hour.	**Medium**	Mobile UX Best Practices	*Proposed*
NFR-P07	Dashboard and analytics performance	Dashboard loads within 4 seconds for datasets up to 10,000 booking records.	**Medium**	Admin	*Proposed*

5.2 RELIABILITY

Reliability requirements ensure that the platform operates continuously and correctly, even in the face of failures. Bus travel, airport transfers, and tour bookings are time-sensitive; any downtime directly impacts passenger travel plans and operator revenue.

ID	Requirement	Measurable Acceptance Criterion	Priority	Source	Status
NFR-R01	System availability	Measured uptime >= 99.5% per calendar month, verified via uptime monitoring tools.	**High**	Client (A9 Express)	*Proposed*
NFR-R02	Server failure handling	Automated failover activates within 60 seconds of any single node failure with no data loss.	**High**	Cloud Infrastructure (AWS)	*Proposed*
NFR-R03	Backup management	Automated daily backups completed successfully; backups retained for 30 days and restorable within 2 hours.	**High**	System Design Team	*Proposed*
NFR-R04	Disaster recovery	RTO <= 2 hours; RPO <= 1 hour of data loss for any unplanned outage.	**High**	System Design Team	*Proposed*
NFR-R05	Failed payment recovery	100% of failed payments are rolled back and users notified within 30 seconds.	**High**	Stripe/PayPal Integration Spec	*Proposed*

NFR-R06	Error monitoring and alerts	Critical errors trigger automated alerts to the dev team within 2 minutes via monitoring service.	**Medium**	System Design Team	*Proposed*
NFR-R07	Maintenance scheduling	All scheduled maintenance occurs between 12 AM – 4 AM with 24-hour advance notification to users.	**Medium**	Operational Policy	*Proposed*

5.3 USABILITY

Usability requirements ensure that all user groups — passengers, operators, and agents — can interact with the platform effectively and efficiently. Survey data (118 respondents) indicates that user adoption resistance is a key risk; therefore, simplicity and accessibility are prioritised.

ID	Requirement	Measurable Acceptance Criterion	Priority	Source	Status
NFR-U01	Easy booking for new users	First-time users complete booking in under 5 minutes in usability testing with no assistance.	**High**	User Survey (118 responses)	*Proposed*
NFR-U02	Multi-language support	Sinhala, Tamil, and English are fully supported; language switchable from any screen.	**High**	Client (A9 Express) / Target Market	*Proposed*
NFR-U03	Accessibility support	Passes WCAG 2.1 Level AA automated and manual audit with zero critical violations.	**Medium**	System Design Team	*Proposed*
NFR-U04	Timely user notifications	SMS and in-app notifications delivered within 60 seconds of booking confirmation,	**High**	Client Requirement	*Proposed*

		cancellation, or schedule change.			
NFR-U05	Easy operator management	Operators complete fleet, route, schedule, and pricing setup within 10 minutes during onboarding testing.	**High**	Client (A9 Express)	*Proposed*
NFR-U06	Clear error messages	All error messages pass user comprehension testing: >= 90% of test users understand the error and next step.	**Medium**	System Design Team	*Proposed*
NFR-U07	Hybrid booking support	Phone-assisted booking option available; confirmed bookings processed within the same system in under 5 minutes.	**Medium**	Target Market Analysis	*Proposed*

5.4 SECURITY

Security requirements protect user data, payment information, and platform integrity. Survey feedback identified payment security (68.6%) and reliability (53.4%) as the top concerns among potential users, making security a high-priority dimension across the system.

ID	Requirement	Measurable Acceptance Criterion	Priority	Source	Status
NFR-S01	Secure data transmission	All endpoints use HTTPS/TLS 1.2+; HTTP requests are redirected to HTTPS. Verified via SSL Labs scan score >= A.	**High**	System Design Team / OWASP	*Proposed*
NFR-S02	Secure authentication	Clerk-managed authentication with OTP/MFA enabled; zero authentication bypass	**High**	Technology Stack (Clerk)	*Proposed*

		incidents in penetration testing.			
NFR-S03	Secure payment processing	PCI-DSS compliant payment flow via Stripe/PayPal; no raw card data stored on platform servers — confirmed by security audit.	**High**	Stripe/PayPal Integration Spec	*Proposed*
NFR-S04	Secure payment processing	RBAC enforced: passengers, operators, agents, and admins cannot access unauthorised resources — validated by role-based test suite.	**High**	System Design Team	*Proposed*
NFR-S05	Fraud detection	Fraud detection flags >= 95% of simulated fraudulent transactions in testing; blocked accounts notified within 30 seconds.	**High**	Risk Analysis	*Proposed*
NFR-S06	Secure password storage	All passwords hashed with bcrypt (cost factor >= 12); verified by code review — plaintext passwords absent from all storage layers.	**High**	Security Best Practices	*Proposed*
NFR-S07	Agent verification	100% of agent accounts require document upload and admin approval; unverified agents cannot process bookings.	**High**	Client Requirement / Risk Analysis	*Proposed*
NFR-S08	Session timeout	Sessions automatically expire after 30 minutes of inactivity on all platforms;	**Medium**	System Design Team	*Proposed*

		user prompted to re-authenticate.			

5.5 SCALABILITY

Scalability requirements ensure the platform can accommodate growth in users, operators, and service types without requiring a full system redesign. The platform is expected to onboard multiple transport operators beyond A9 Express as the business grows.

ID	Requirement	Measurable Acceptance Criterion	Priority	Source	Status
NFR-SC01	Independent service scaling	Each service (booking, payment, notifications) scales independently; verified by load test showing no cascading failures.	**High**	System Design Team	*Proposed*
NFR-SC02	Automatic scaling during traffic spikes	AWS Auto Scaling provisions additional capacity within 3 minutes of CPU/load threshold breach with no downtime.	**High**	Cloud Infrastructure (AWS)	*Proposed*
NFR-SC03	Large database handling	Query response time remains under 2 seconds with 1 million booking records in the database.	**High**	System Design Team	*Proposed*
NFR-SC04	Easy onboarding of new operators	A new transport operator or service type can be onboarded within 2 developer-days without core architecture changes.	**Medium**	Client (A9 Express) Growth Plan	*Proposed*
NFR-SC05	Data caching efficiency	Cache hit rate >= 70% for routes, schedules, and seat availability queries —	**Medium**	System Design Team	*Proposed*

		measured via Redis monitoring.			
NFR-SC06	High API request handling	API sustains >= 1,000 requests/minute per operator with < 1% error rate under load testing.	**Medium**	System Design Team	*Proposed*

5.6 MAINTAINABILITY

Maintainability requirements ensure that the system can be efficiently updated, debugged, and extended over its operational lifetime. The team has a 4-month development timeline followed by ongoing maintenance; a clean, documented codebase is essential for long-term project health.

ID	Requirement	Measurable Acceptance Criterion	Priority	Source	Status
NFR-M01	Consistent and documented codebase	ESL int/Prettier pass with zero errors on CI; >= 80% of functions have inline documentation comments.	**High**	System Design Team	*Proposed*
NFR-M02	Automated testing and deployment	CI/CD pipeline (GitHub Actions) executes all tests and deploys to staging on every pull request merge with zero manual steps.	**High**	Technology Stack (GitHub)	*Proposed*
NFR-M03	Automated test coverage	Unit and integration test coverage >= 70% across booking, payment, and authentication modules — measured by Jest coverage report.	**Medium**	System Design Team	*Proposed*
NFR-M04	API documentation	OpenAPI/Swagger documentation published and up to date; new	**Medium**	System Design Team	*Proposed*

		developers complete onboarding within 5 working days.			
NFR-M05	Logging and audit support	Structured JSON logs retained for >= 90 days; log search returns results for any event within 10 seconds.	**Medium**	Operational Policy	*Proposed*
NFR-M06	Dependency maintenance	Dependency audit run quarterly; critical/high severity vulnerabilities patched within 14 days of disclosure.	**Low**	System Design Team	*Proposed*

5.7 Portability

Portability requirements ensure that the A9 Express platform can run across different devices, browsers, and deployment environments. Given that the target market spans mobile-first university students and working professionals using varied devices, broad device and browser compatibility is essential.

ID	Requirement	Measurable Acceptance Criterion	Priority	Source	Status
NFR-PO01	Cross-browser compatibility	Fully functional on Chrome, Firefox, Safari, and Edge (latest 2 versions); zero critical browser-specific bugs in QA.	**High**	System Design Team	*Proposed*
NFR-PO02	Cross-platform mobile support	Single React Native codebase deployed to Android (v8.0+) and iOS (v13+); all core features functional on both platforms.	**High**	Technology Stack	*Proposed*

NFR-PO03	Responsive web design	Layout renders correctly and all features are usable at screen widths from 360px to 1920px — verified by device emulation testing.	**High**	System Design Team / Target Market	*Proposed*
NFR-PO04	Cloud deployment portability	Services containerised with Docker; deployment to a different cloud provider (e.g., GCP) achievable within 1 working day.	**Medium**	System Design Team	*Proposed*
NFR-PO05	Database migration support	Schema documented and migration scripts prepared; estimated migration effort to PostgreSQL <= 2 weeks.	**Low**	System Design Team	*Proposed*
NFR-PO06	External configuration management	All secrets and environment variables stored in .env files / secret manager; zero hardcoded credentials in the codebase confirmed by code scan.	**Medium**	DevOps Best Practices	*Proposed*

6. Chapter

BUSINESS RULES, CONSTRAINTS AND ASSUMPTIONS

6.1 Business Rules

Table 5-1: Business Rules

ID	Rule Description
BR001	User passwords shall comply with defined security and complexity requirements, and passwords shall be stored using secure hashing mechanisms.
BR002	Each customer shall be assigned a unique customer identifier within the system.
BR003	User accounts shall be temporarily locked after five consecutive failed login attempts.
BR004	Customers shall be permitted to book buses, airport transfers, and tour packages only after successful authentication.
BR005	Operators shall be permitted to manage routes, schedules, and pricing but shall not be permitted to modify system-level settings.
BR006	Drivers shall only be permitted to access assigned trips and schedules.
BR007	Administrative users shall be responsible for managing users, bookings, routes, pricing, and system reports.
BR008	Password reset operations shall be performed through a secure email verification process.
BR009	A customer shall not be allowed to reserve a seat that has already been booked by another customer.
BR010	Incomplete bookings shall be automatically cancelled if payment or confirmation is not completed within 15 minutes.
BR011	Customers shall be allowed to cancel bookings only within the permitted cancellation period defined by the company.
BR012	Real-time booking confirmation notifications shall be sent to customers through email or SMS after successful booking completion.
BR013	Ticket pricing for bus routes, airport transfers, and tour packages shall be configurable by the Admin.
BR014	Tour package pricing shall be automatically calculated based on selected destinations, vehicle type, and duration.

6.2 Technology Constraints

Table 5-2: Technology Constraints

ID	Constraint
CON001	The system shall be implemented as a web-based application accessible through modern web browsers such as Chrome, Edge, Firefox, and Safari.
CON002	The system frontend shall be developed using React.js and the backend shall be developed using Node.js.
CON003	The system database shall be implemented using MySQL.
CON004	The system shall support responsive user interfaces for desktop and mobile devices.
CON005	Internet connectivity shall be required for accessing online booking services and notifications.

6.3 Legal Constraints

Table 5-3: Legal Constraints

ID	Constraint
CON006	Customer personal information shall be handled according to applicable data privacy and protection regulations.
CON007	Access to booking and customer information shall be restricted to authorized users only.
CON008	Payment and transaction data shall be securely stored and protected from unauthorized access.
CON009	Customer booking records shall be retained according to company and legal retention policies.

6.4 Operational Constraints

Table 5-4: Operational Constraints

ID	Constraint
CON010	The system shall support multiple transport services including bus booking, airport transfers, and tour package management.
CON011	The system shall maintain operational continuity under moderate internet connectivity conditions.
CON012	Real-time seat availability updates shall be supported during high booking activity periods.
CON013	Booking and payment confirmation notifications shall depend on third-party email or SMS gateway availability.

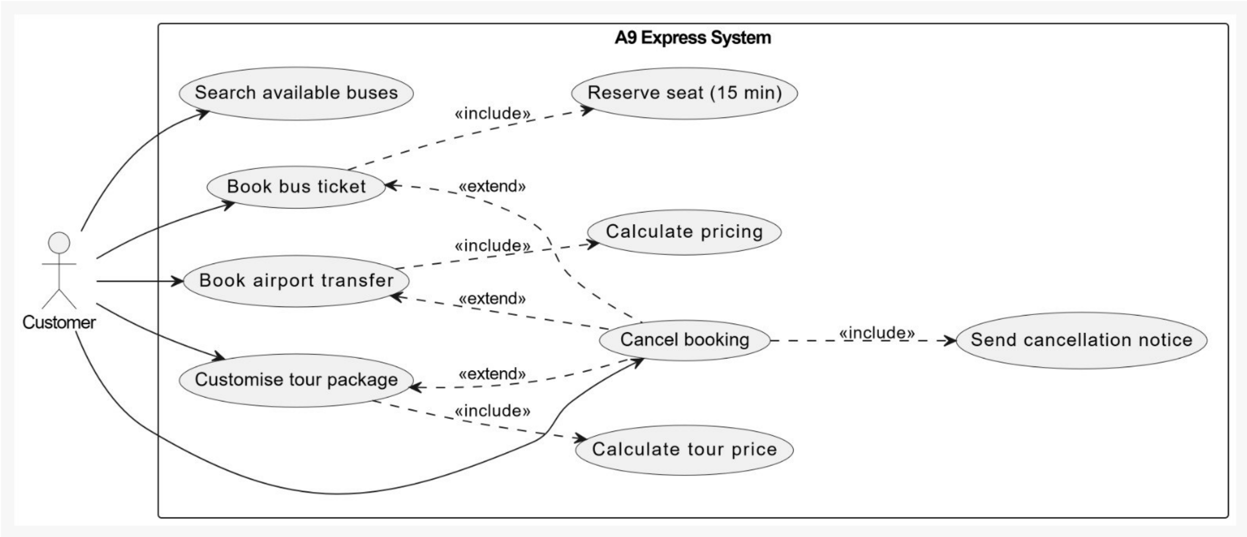
6.5 Assumptions

Table 5-5: System Assumptions

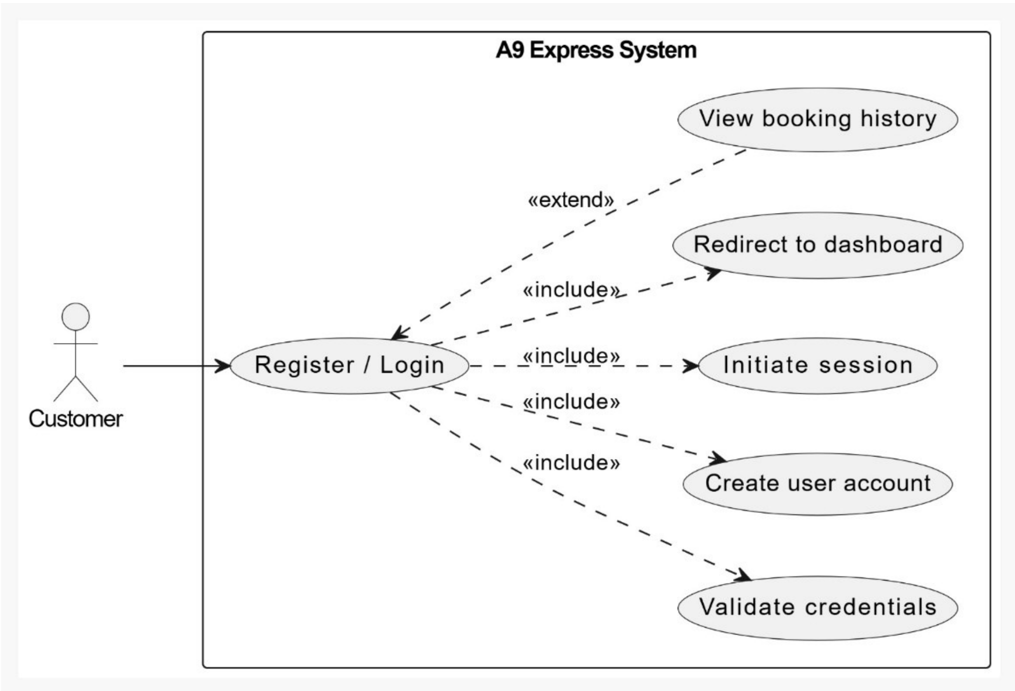
ID	Assumption Description
ASM001	Customers and staff are assumed to possess basic computer and smartphone literacy to operate the system effectively.
ASM002	Reliable internet connectivity is assumed to be available for accessing the online booking platform.
ASM003	Transport operators are assumed to provide accurate route, schedule, and pricing information to the system.
ASM004	Administrators are assumed to regularly maintain and update booking schedules, vehicles, and driver details.
ASM005	Secure hosting infrastructure and regular database backups are assumed to be available for the system.
ASM006	Email and SMS notification services are assumed to be operational and accessible when sending booking confirmations and alerts.
ASM007	Customers are assumed to provide valid personal and payment information during registration and booking processes.

7. CHAPTER
USE CASE DIAGRAMS

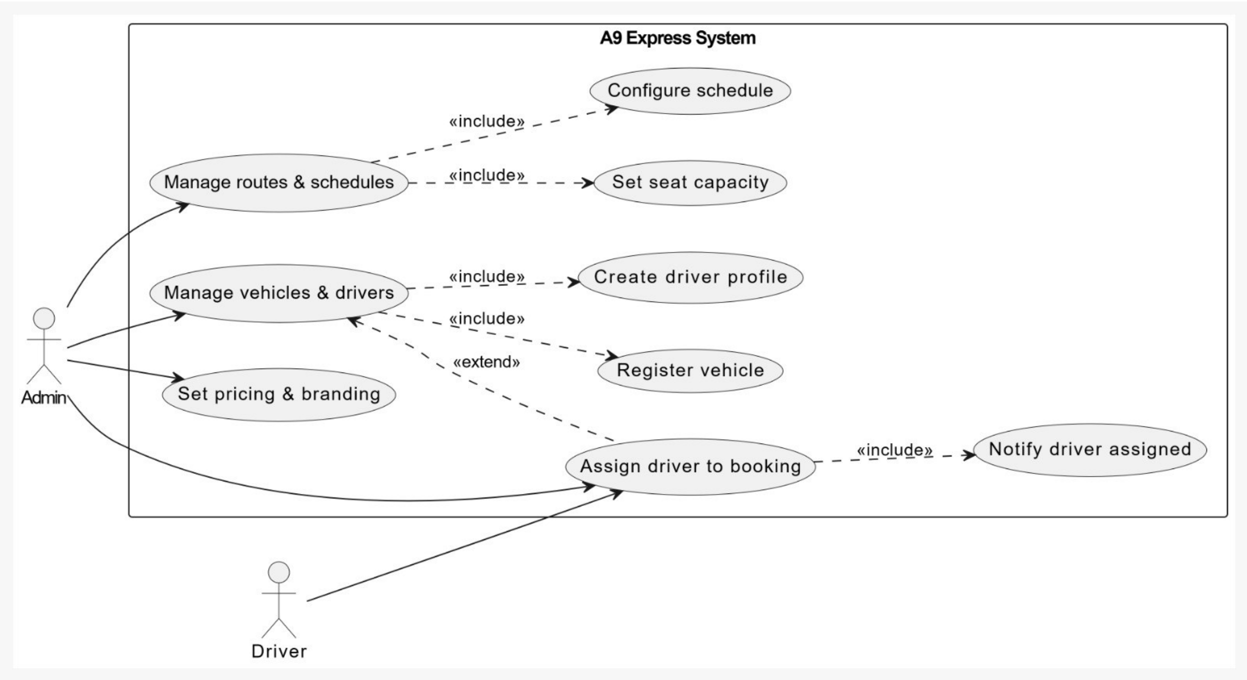
7.1 Coustomer Booking Flow



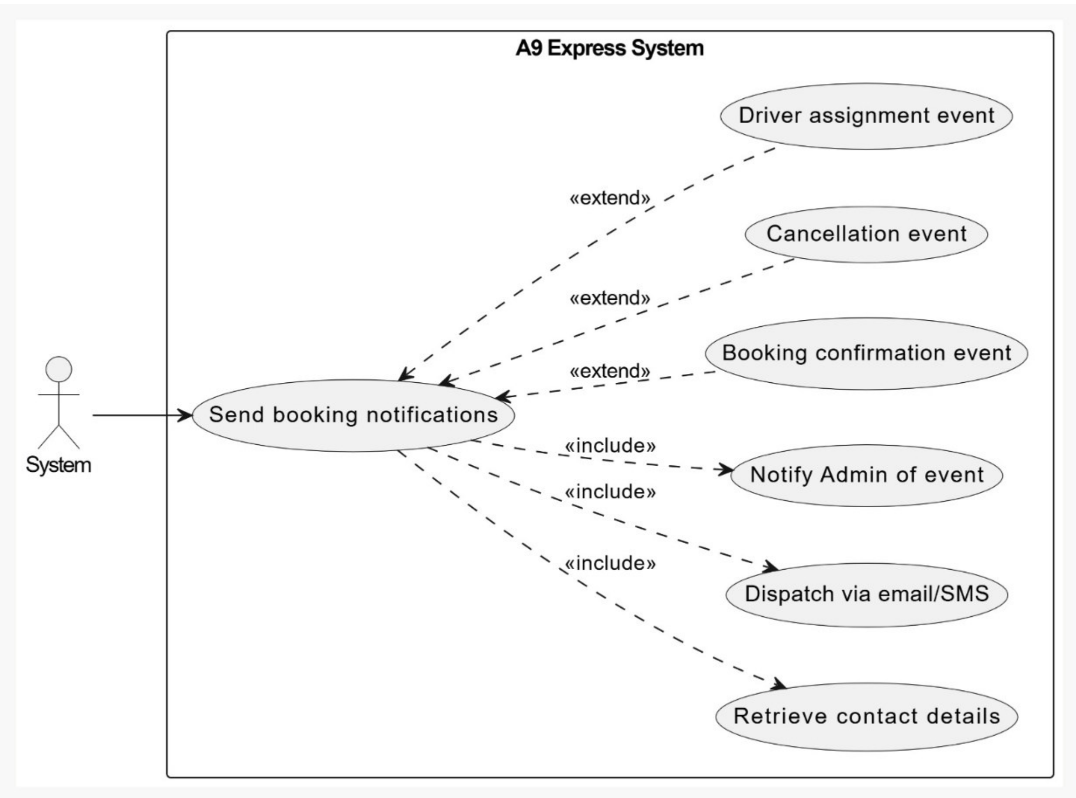
7.2 Authentication and Account Access



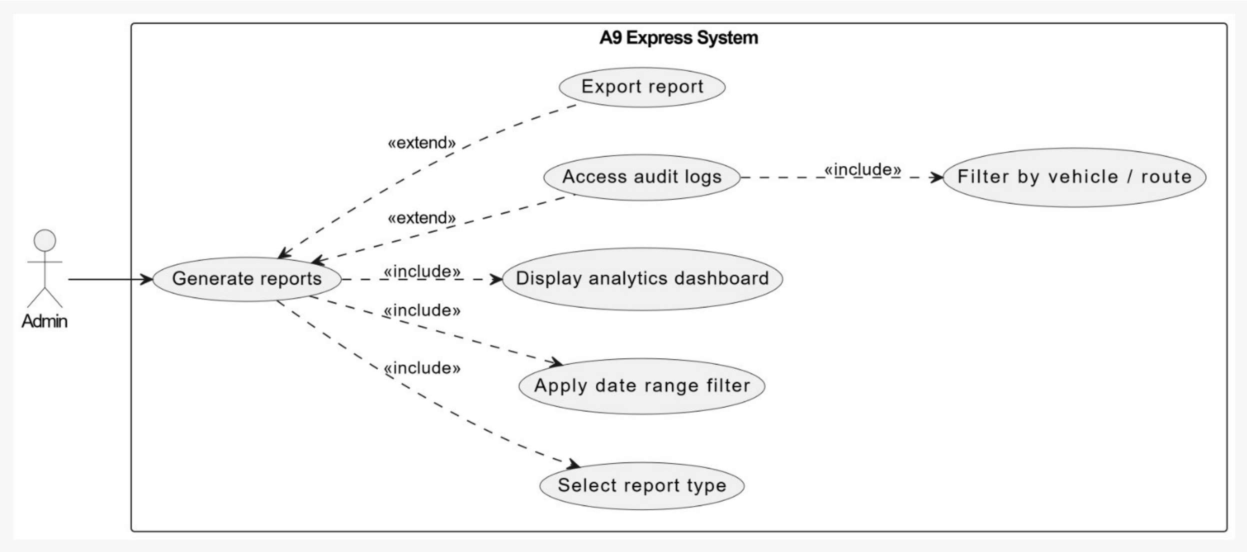
7.3 Admin Route & Vehicle Management



7.4 Notification and System Events



7.5 Admin Reporting and Analytics



8. CHAPTER
USE CASE DESCRIPTIONS

8.1 Diagram- 1 Customer Booking Flow

8.1.1 UC-001: Search Available Buses

Use Case ID	UC-001
Use Case Name	Search Available Buses
Actor(s)	Customer
Description	Customer searches for available buses by selecting origin, destination, and travel date.
Preconditions	Customer is logged in. Bus routes and schedules have been configured by Admin.
Postconditions – Success	Customer views available buses and proceeds to booking.
Main Flow	1. Customer navigates to the bus search section. 2. Customer selects origin, destination, and travel date. 3. System queries available buses. 4. System displays list of available buses with seat availability. 5. Customer selects a preferred bus.
«include»	—
«extend»	—

8.1.2 UC-002: Book Bus Ticket

Use Case ID	UC-002
Use Case Name	Book Bus Ticket
Actor(s)	Customer, System
Description	Customer selects and confirms a seat reservation on an available bus route.
Preconditions	Customer is logged in. Customer has searched and selected an available bus. Seats are available.
Postconditions – Success	Seat(s) reserved. Booking confirmation sent. Seat availability updated.
Postconditions – Failure	Booking not confirmed. Seat reservation released after timeout.
Main Flow	1. Customer selects one or more available seats. 2. System temporarily reserves seat(s) for 15 minutes. 3. Customer reviews and confirms booking. 4. System processes and confirms the booking. 5. Confirmation sent via email or SMS.
Alternative Flow 1	At step 1: If no seats are available, system displays a 'Sold Out' message and suggests alternative times or routes.
Alternative Flow 2	At step 3: If customer does not confirm within 15 minutes, system releases the temporary reservation.
Business Rules	BR-01: Seat reservations expire after 15 minutes. BR-02: Booking confirmation must be sent within 2 minutes of confirmation.
«include»	Reserve seat (15 min)
«extend»	Cancel booking

8.1.3 UC-003: Book Airport Transfer

Use Case ID	UC-003
Use Case Name	Book Airport Transfer
Actor(s)	Customer, Admin
Description	Customer books an airport pickup or drop-off by providing travel details.
Preconditions	Customer is logged in. Airport transfer services are available.
Postconditions – Success	Transfer booked and confirmed. Customer receives notification.
Main Flow	1. Customer navigates to airport transfer section. 2. Customer provides date, time, location, and flight details. 3. System displays available vehicles and estimated pricing. 4. Customer selects vehicle and confirms booking. 5. System generates confirmation and notifies customer.
«include»	Calculate pricing
«extend»	Cancel booking

8.1.4 UC-004: Customise Tour Package

Use Case ID	UC-004
Use Case Name	Customise Tour Package
Actor(s)	Customer, System
Description	Customer creates a personalised Sri Lankan tour package by selecting tourist destinations.
Preconditions	Customer is logged in. Tourist destinations and pricing configured by Admin.

Postconditions – Success	Tour package created and confirmed. Booking visible in customer history.
Main Flow	1. Customer navigates to tour package section. 2. Customer selects destinations from available list. 3. Customer selects vehicle type and specifies duration. 4. System calculates total package price. 5. Customer reviews, saves, and submits the tour booking. 6. System generates confirmation and notifies customer.
«include»	Calculate tour price
«extend»	Cancel booking

8.1.5 UC-005: Cancel Booking

Use Case ID	UC-005
Use Case Name	Cancel Booking
Actor(s)	Customer, System
Description	Customer cancels an existing bus, airport transfer, or tour booking within the permitted cancellation window.
Preconditions	Customer is logged in. Customer has an active confirmed booking. Cancellation is within the permitted time window.
Postconditions – Success	Booking status updated to Cancelled. Notifications sent. Availability restored.
Postconditions – Failure	Cancellation not permitted. Customer notified of cancellation window violation.
Main Flow	1. Customer navigates to booking history. 2. Customer selects the booking to cancel. 3. System validates cancellation is within permitted window. 4. Customer confirms cancellation.

	5. System updates booking status to Cancelled. 6. System sends cancellation confirmation to customer and Admin. 7. System restores seat or vehicle availability.
Alternative Flow 1	At step 3: If cancellation window has expired, system displays an error message and denies the cancellation request.
Business Rules	BR-03: Cancellations must be requested at least 2 hours before scheduled departure.
«include»	Send cancellation notice
«extend»	Triggered from: Book bus ticket, Book airport transfer, Customise tour package

8.2 Diagram 2 – Authentication & Account Access

8.2.1 UC-006: Register / Login

Use Case ID	UC-006
Use Case Name	Register / Login
Actor(s)	Customer
Description	A new customer registers an account or an existing customer authenticates to access the platform.
Preconditions	User has access to the A9 Express platform. User has a valid email address.
Postconditions – Success	Customer account created or authenticated. Session initiated. Customer redirected to dashboard.
Postconditions – Failure	Account not created or login denied. Error message displayed.

Main Flow	1. New customer navigates to registration page and provides name, email, phone, and password. 2. System validates details and creates the account. 3. Existing customer enters valid email and password. 4. System authenticates the user. 5. System initiates session and redirects to dashboard.
Alternative Flow 1	At step 1 (Registration): If email is already registered, system notifies user and prompts login instead.
Alternative Flow 2	At step 3 (Login): If credentials are invalid after 3 attempts, account is temporarily locked for 15 minutes.
Business Rules	BR-06: Passwords must be at least 8 characters with one uppercase letter and one number. BR-07: Account lockout after 3 failed login attempts.
«include»	Validate credentials, Create user account, Initiate session, Redirect to dashboard
«extend»	View booking history

8.2.2 UC-007: Validate Credentials

Use Case ID	UC-007
Use Case Name	Validate Credentials
Actor(s)	System
Description	System validates the email and password provided by the customer during login.
Preconditions	Customer has submitted login credentials.
Postconditions – Success	Credentials validated; authentication proceeds.

Postconditions – Failure	Invalid credentials; error message returned to customer.
Main Flow	1. System receives email and password input. 2. System checks credentials against stored records. 3. If valid, authentication proceeds. 4. If invalid, system returns an error message.
«include»	—
«extend»	—

8.2.3 UC-008: Initiate Session

Use Case ID	UC-008
Use Case Name	Initiate Session
Actor(s)	System
Description	System creates an authenticated session for the customer after successful login.
Preconditions	Credentials have been successfully validated.
Postconditions – Success	Active session created. Customer has access to the platform.
Main Flow	1. System generates a session token for the authenticated user. 2. Session is stored and associated with the customer account. 3. Customer is granted access to protected platform features.
«include»	—
«extend»	—

8.2.4 UC-009: View Booking History

Use Case ID	UC-009
Use Case Name	View Booking History
Actor(s)	Customer
Description	Customer views all current and past bookings from their dashboard after login.
Preconditions	Customer is logged in.
Postconditions – Success	Customer views complete booking history. Booking details displayed accurately.
Main Flow	1. Customer navigates to booking history section on dashboard. 2. System retrieves all current and past bookings. 3. Customer views booking details including service type, date, time, and payment status. 4. Customer selects a specific booking to view full details.
«include»	—
«extend»	Extends Register / Login

8.3 Diagram -3 Admin Route & Vehicle Management

8.3.1 UC-010: Manage Routes & Schedules

Use Case ID	UC-010
Use Case Name	Manage Routes & Schedules
Actor(s)	Admin
Description	Admin creates, updates, and deactivates bus routes and configures schedules.
Preconditions	Admin is logged in with appropriate permissions.

Postconditions – Success	New route and schedule are active and bookable. Seat availability initialised.
Main Flow	1. Admin navigates to route management section. 2. Admin creates a new route by defining origin, destination, and stops. 3. Admin configures schedules including departure times and days of operation. 4. Admin sets ticket pricing per route. 5. System saves and makes the route available for customer booking.
«include»	Configure schedule, Set seat capacity
«extend»	—

8.3.2 UC-011: Manage Vehicles & Drivers

Use Case ID	UC-011
Use Case Name	Manage Vehicles & Drivers
Actor(s)	Admin
Description	Admin manages the fleet of vehicles and driver profiles on the platform.
Preconditions	Admin is logged in with appropriate permissions.
Postconditions – Success	Vehicle and driver profiles saved and available for assignment.
Main Flow	1. Admin navigates to vehicle management section. 2. Admin adds a new vehicle with type, capacity, and registration details. 3. Admin navigates to driver management section. 4. Admin creates or updates a driver profile with personal details and licence information. 5. Admin sets driver status as active or inactive. 6. System saves vehicle and driver information.

«include»	Register vehicle, Create driver profile
«extend»	Assign driver to booking

8.3.3 UC-012: Assign Driver to Booking

Use Case ID	UC-012
Use Case Name	Assign Driver to Booking
Actor(s)	Admin, Driver
Description	Admin assigns a driver and vehicle to a confirmed airport transfer or tour booking.
Preconditions	Admin is logged in. A confirmed booking exists requiring a driver. An available driver and vehicle exist.
Postconditions – Success	Driver and vehicle assigned to booking. Vehicle availability updated. Driver notified.
Main Flow	1. Admin views confirmed bookings requiring driver assignment. 2. Admin selects a booking to assign. 3. Admin selects an available driver and vehicle. 4. System assigns driver and vehicle to the booking. 5. System updates vehicle availability. 6. Driver is notified of the assignment.
«include»	Notify driver assigned
«extend»	Extends Manage vehicles & drivers

8.4 Diagram-4 Notification & System Events

8.4.1 UC-013: Send Booking Notifications

Use Case ID	UC-013
Use Case Name	Send Booking Notifications
Actor(s)	System
Description	System automatically sends real-time notifications to customers and Admin on booking events.
Preconditions	A booking, cancellation, or assignment event has occurred. Customer contact details are available.
Postconditions – Success	Customer receives booking or cancellation confirmation. Admin notified of relevant events.
Main Flow	1. System detects a booking confirmation, cancellation, or driver assignment event. 2. System retrieves customer contact details. 3. System generates the appropriate notification message. 4. System dispatches notification via email or SMS. 5. System notifies Admin of relevant booking events.
«include»	Retrieve contact details, Dispatch via email/SMS, Notify Admin of event
«extend»	Booking confirmation event, Cancellation event, Driver assignment event

8.4.2 UC-014: Booking Confirmation Event

Use Case ID	UC-014
Use Case Name	Booking Confirmation Event
Actor(s)	System

Description	System triggers a notification when a new booking is successfully confirmed.
Preconditions	A booking has been successfully created and confirmed.
Postconditions – Success	Customer notified of confirmed booking.
Main Flow	1. System detects booking status set to Confirmed. 2. System triggers Send booking notifications use case. 3. Customer receives booking confirmation with details.
«include»	—
«extend»	Extends Send booking notifications

8.4.3 UC-015: Cancellation Event

Use Case ID	UC-015
Use Case Name	Cancellation Event
Actor(s)	System
Description	System triggers a notification when a booking is cancelled by the customer.
Preconditions	A booking has been successfully cancelled.
Postconditions – Success	Customer and Admin notified of cancellation.
Main Flow	1. System detects booking status updated to Cancelled. 2. System triggers Send booking notifications use case. 3. Customer and Admin receive cancellation confirmation.
«include»	—
«extend»	Extends Send booking notifications

8.4.4 UC-016: Driver Assignment Event

Use Case ID	UC-016
Use Case Name	Driver Assignment Event
Actor(s)	System
Description	System triggers a notification when a driver is assigned to a confirmed booking.
Preconditions	A driver has been successfully assigned to a booking by Admin.
Postconditions – Success	Driver notified of booking assignment.
Main Flow	1. System detects driver assignment to a confirmed booking. 2. System triggers Send booking notifications use case. 3. Driver receives assignment notification with booking details.
«include»	—
«extend»	Extends Send booking notifications

8.5 Diagram-5 Admin Reporting & Analytics

8.5.1 UC-017: Generate Reports

Use Case ID	UC-017
Use Case Name	Generate Reports
Actor(s)	Admin
Description	Admin generates operational reports and views analytics on bookings and revenue.
Preconditions	Admin is logged in. Booking and operational data exists in the system.
Postconditions – Success	Report generated and displayed to Admin.

Main Flow	1. Admin navigates to reporting and analytics section. 2. Admin selects report type (bookings, revenue, vehicle utilisation). 3. Admin specifies date range and filters. 4. System generates and displays the report. 5. Admin views analytics dashboard with booking trends and revenue summaries.
«include»	Select report type, Apply date range filter, Display analytics dashboard
«extend»	Access audit logs, Export report

8.5.2 UC-018: Access Audit Logs

Use Case ID	UC-018
Use Case Name	Access Audit Logs
Actor(s)	Admin
Description	Admin optionally accesses detailed audit logs for deeper review of system activity.
Preconditions	Admin has generated a report. Admin requires detailed activity tracking.
Postconditions – Success	Audit logs displayed and accessible for review.
Main Flow	1. Admin selects the audit log option from the reporting section. 2. System retrieves audit log entries. 3. Admin applies filter by vehicle or route. 4. System displays filtered audit log results.
«include»	Filter by vehicle / route
«extend»	Extends Generate reports

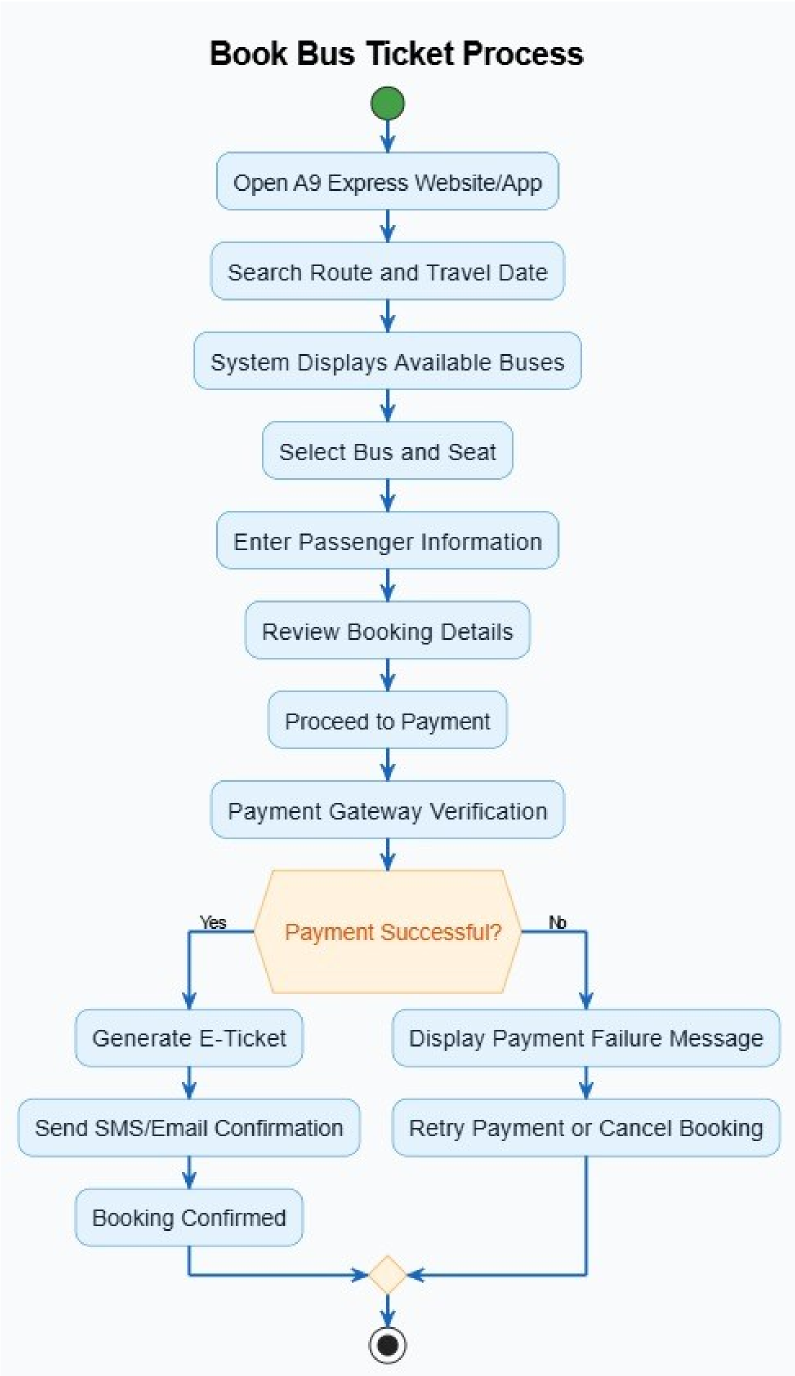
8.5.3 UC-019: Export Report

Use Case ID	UC-019
Use Case Name	Export Report
Actor(s)	Admin
Description	Admin optionally exports a generated report for external use or record keeping.
Preconditions	Admin has generated a report.
Postconditions – Success	Report exported and downloaded by Admin.
Main Flow	1. Admin selects the export option after viewing a report. 2. Admin chooses the preferred export format. 3. System generates the export file. 4. Admin downloads the file.
«include»	—
«extend»	Extends Generate reports

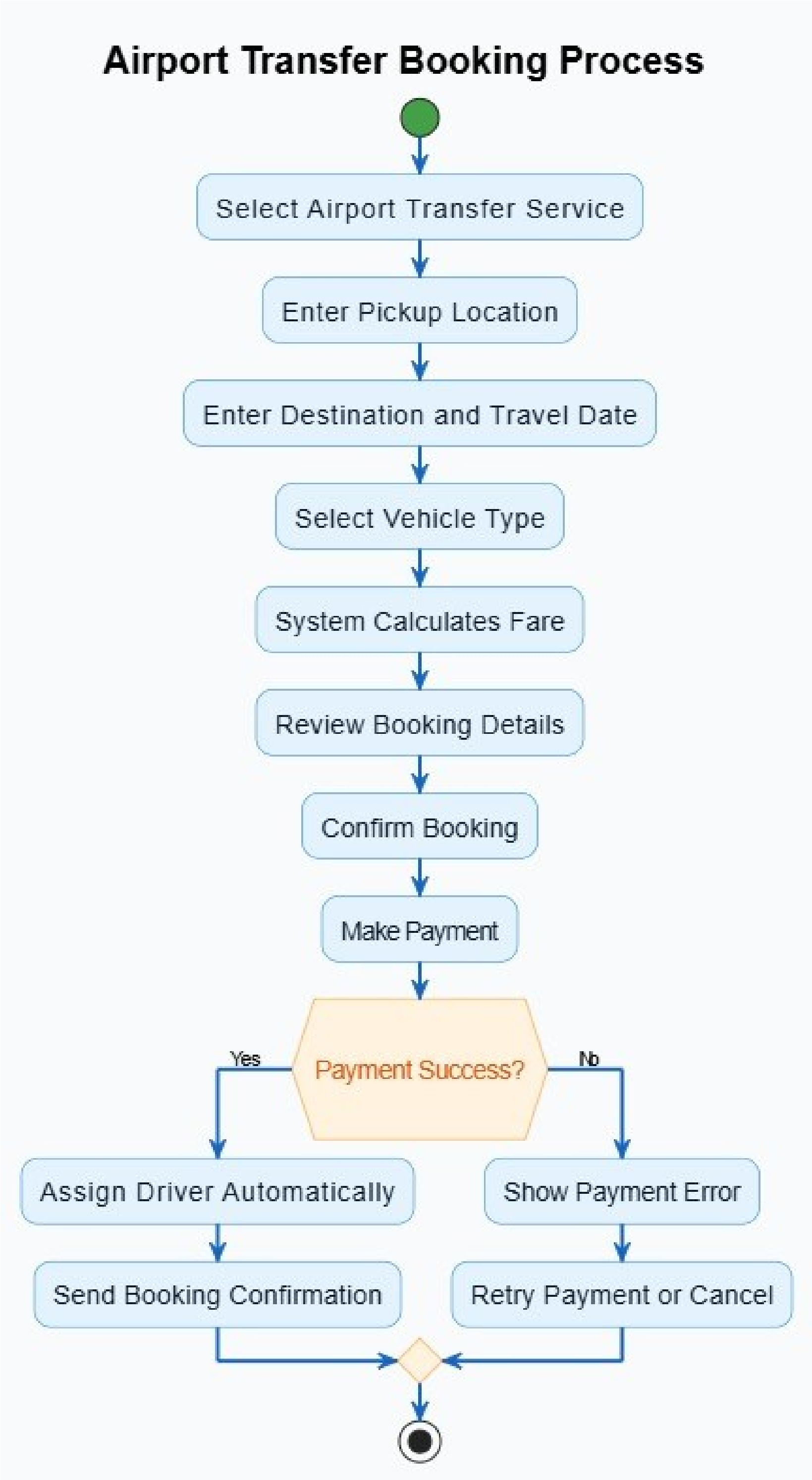
9. CHAPTER
ACTIVITY & SEQUENCE DIAGRAMS

9.1 ACTIVITY DIAGRAMS

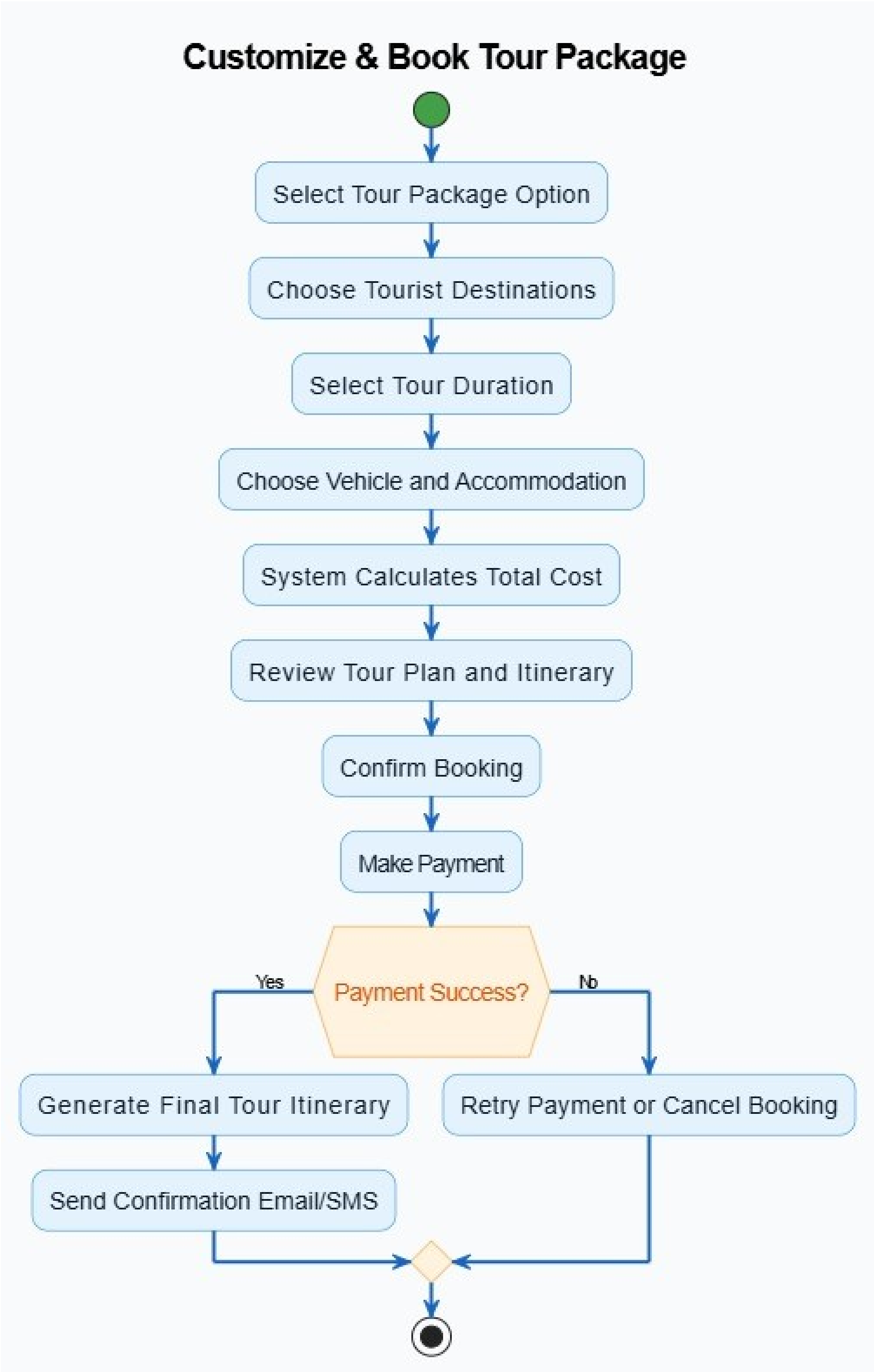
9.1.1 Activity Diagram – Book Bus Ticket



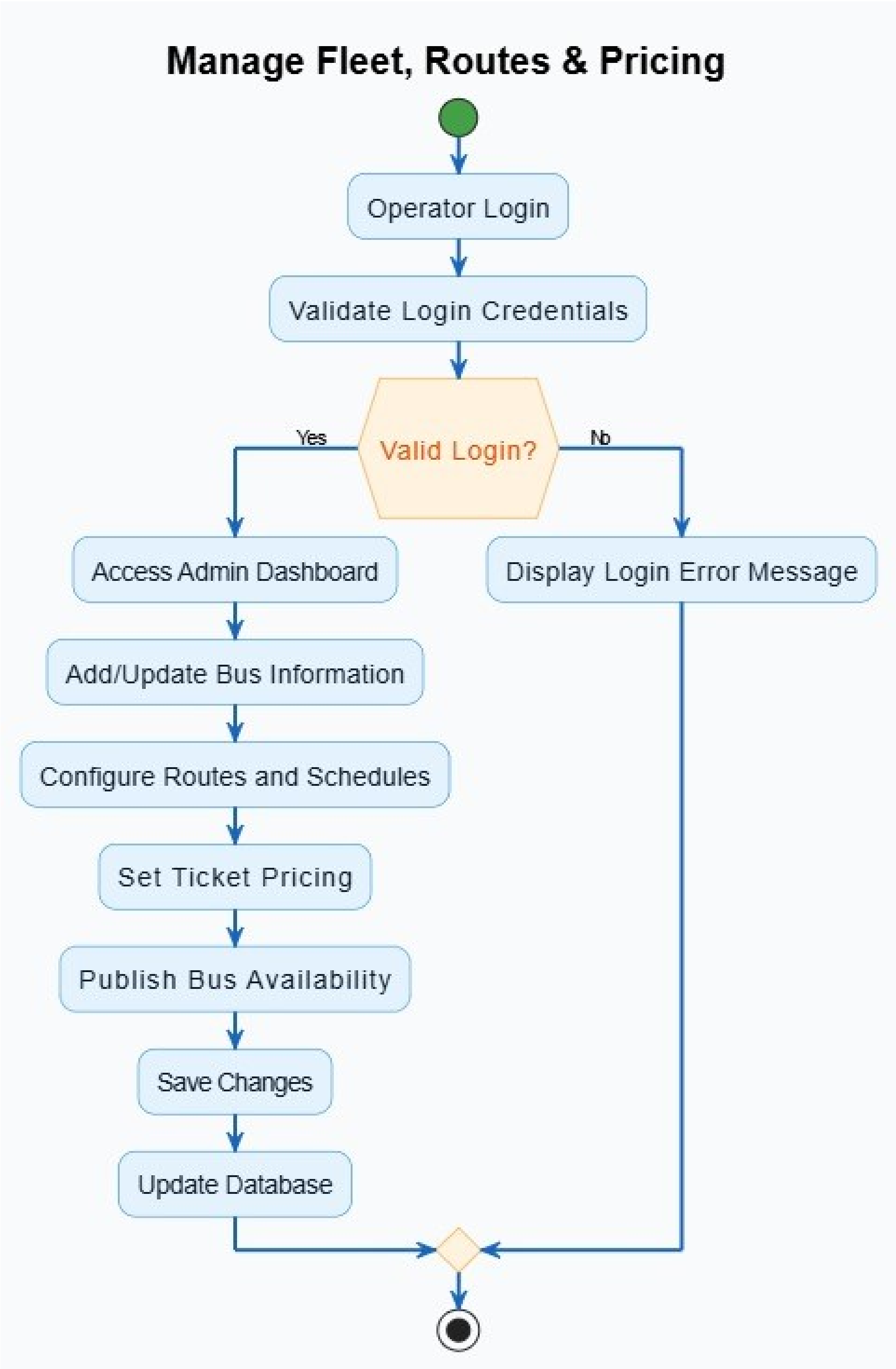
9.1.2 Activity Diagram – Airport Transfer Booking



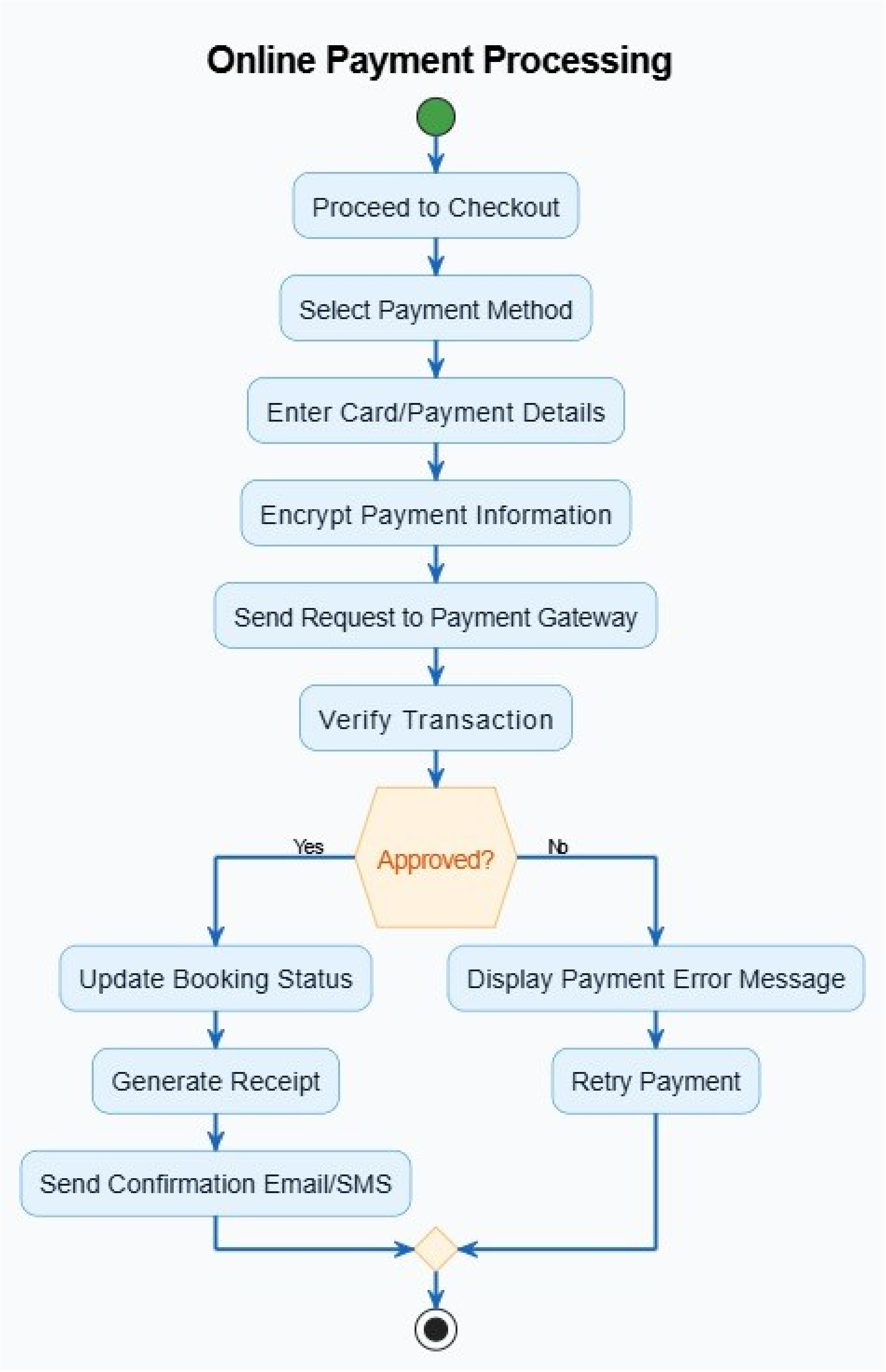
9.1.3 Activity Diagram – Customize & Book Tour Package



9.1.4 Activity Diagram – Manage Fleet, Routes & Pricing

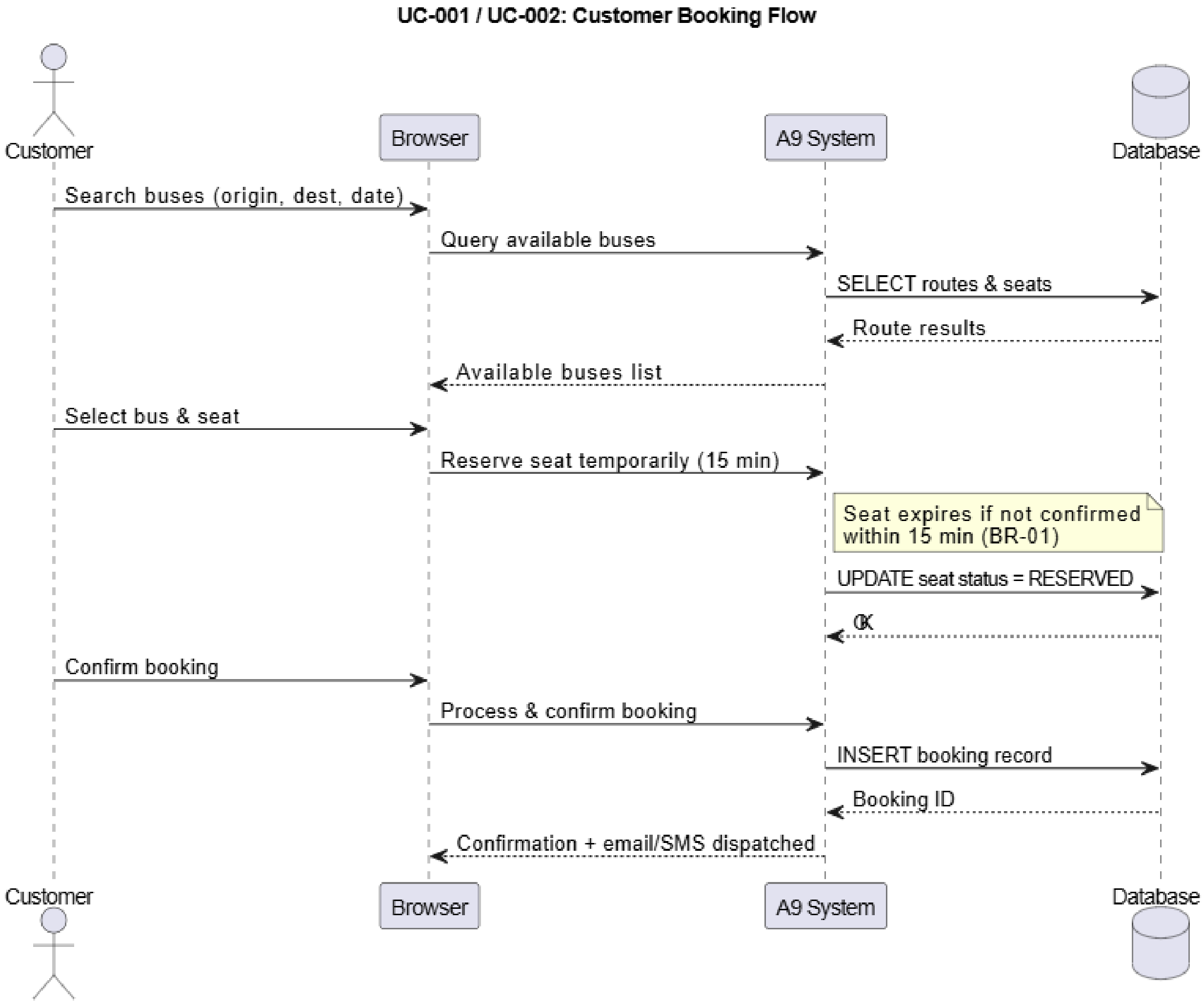


9.1.5 Activity Diagram – Online Payment Processing

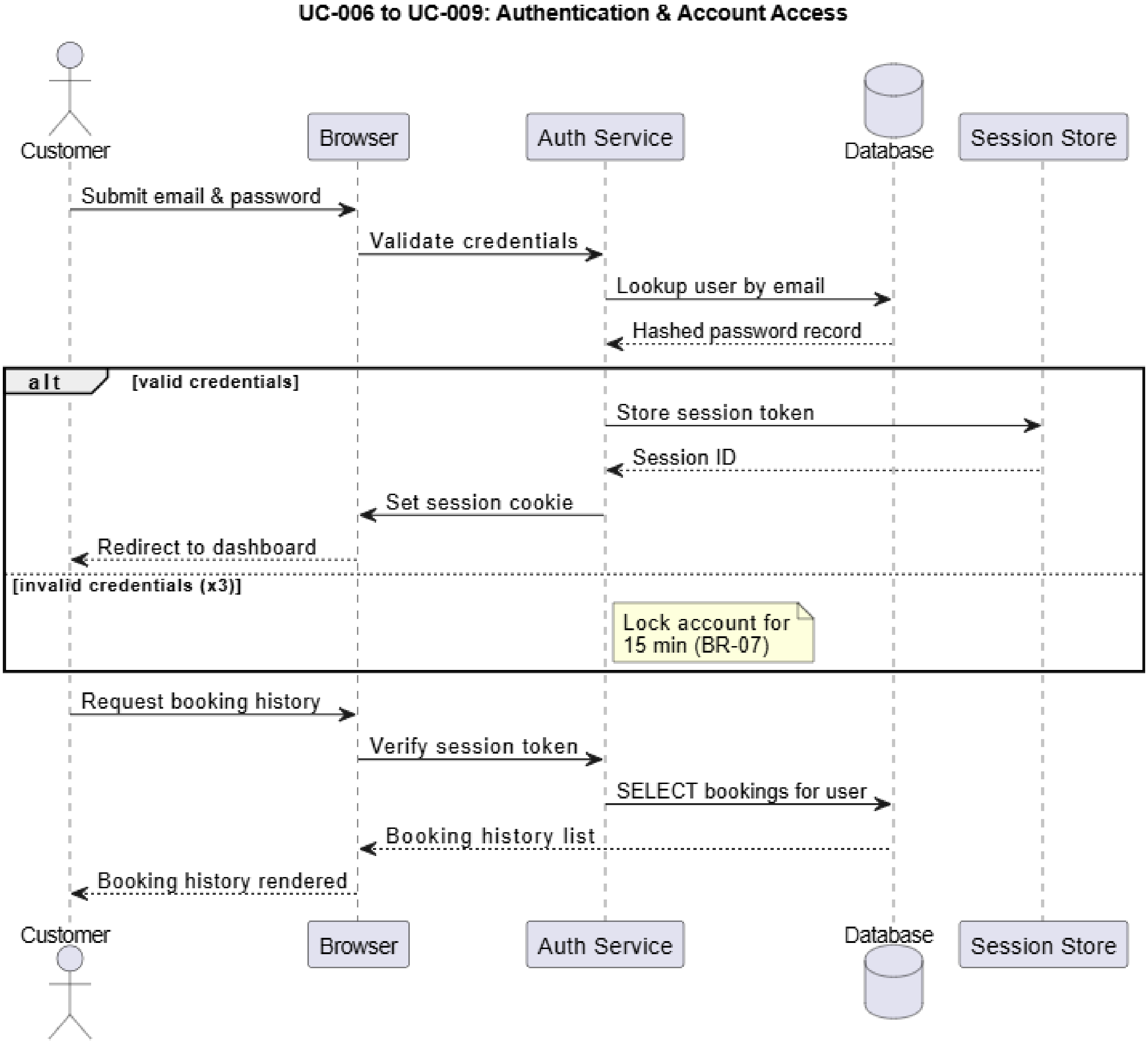


9.2 SEQUENCE DIAGRAMS

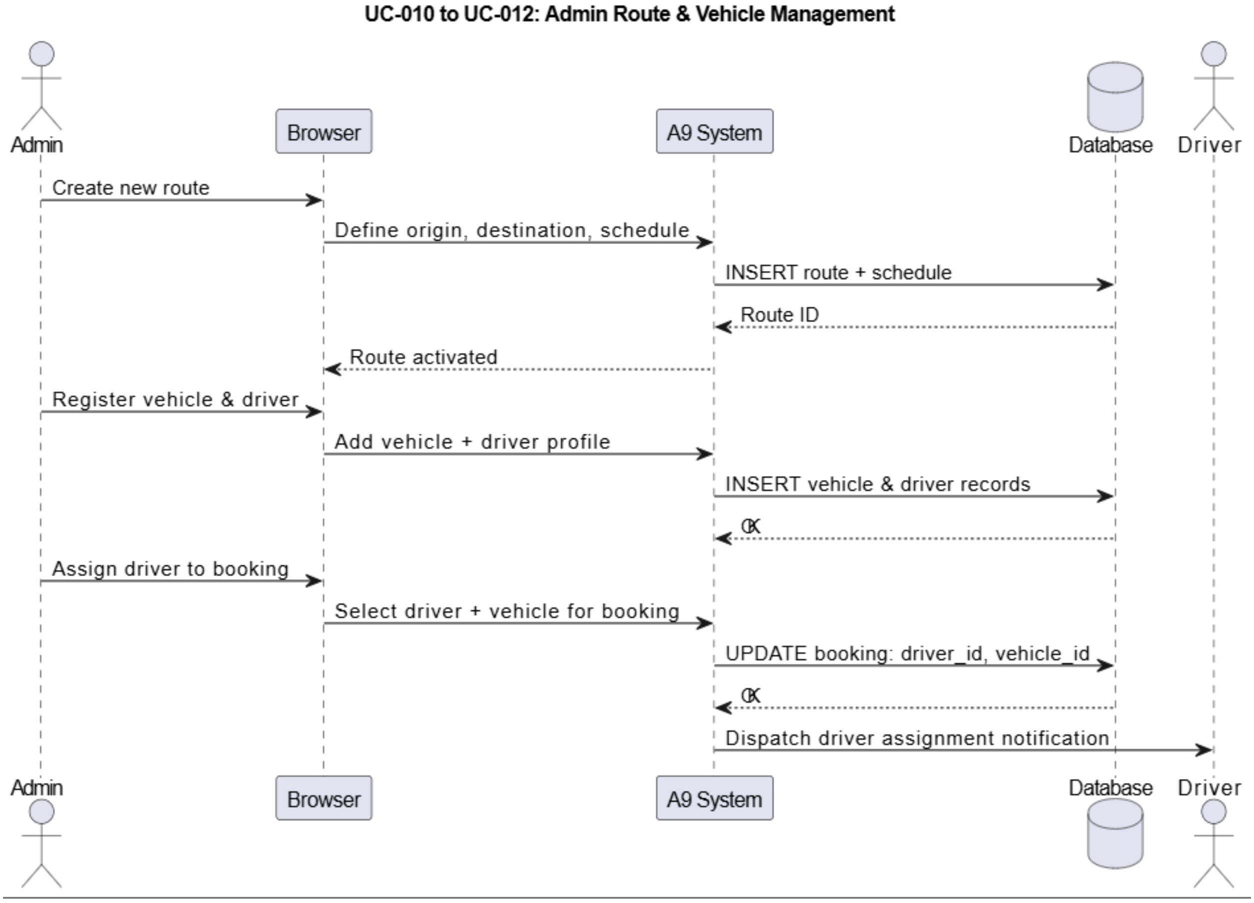
9.2.1 Customer Booking Flow



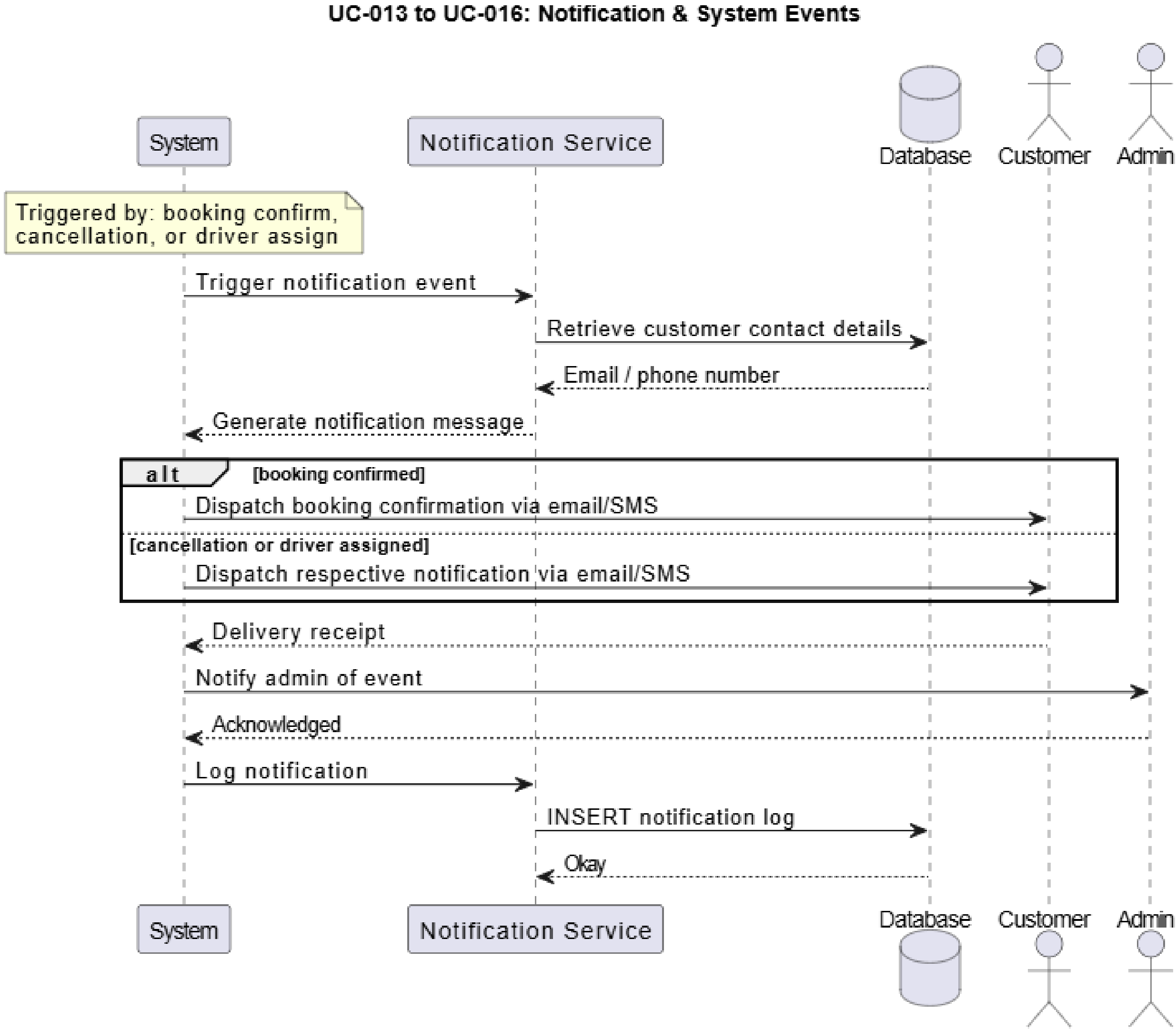
9.2.2 Authentication & Account Access



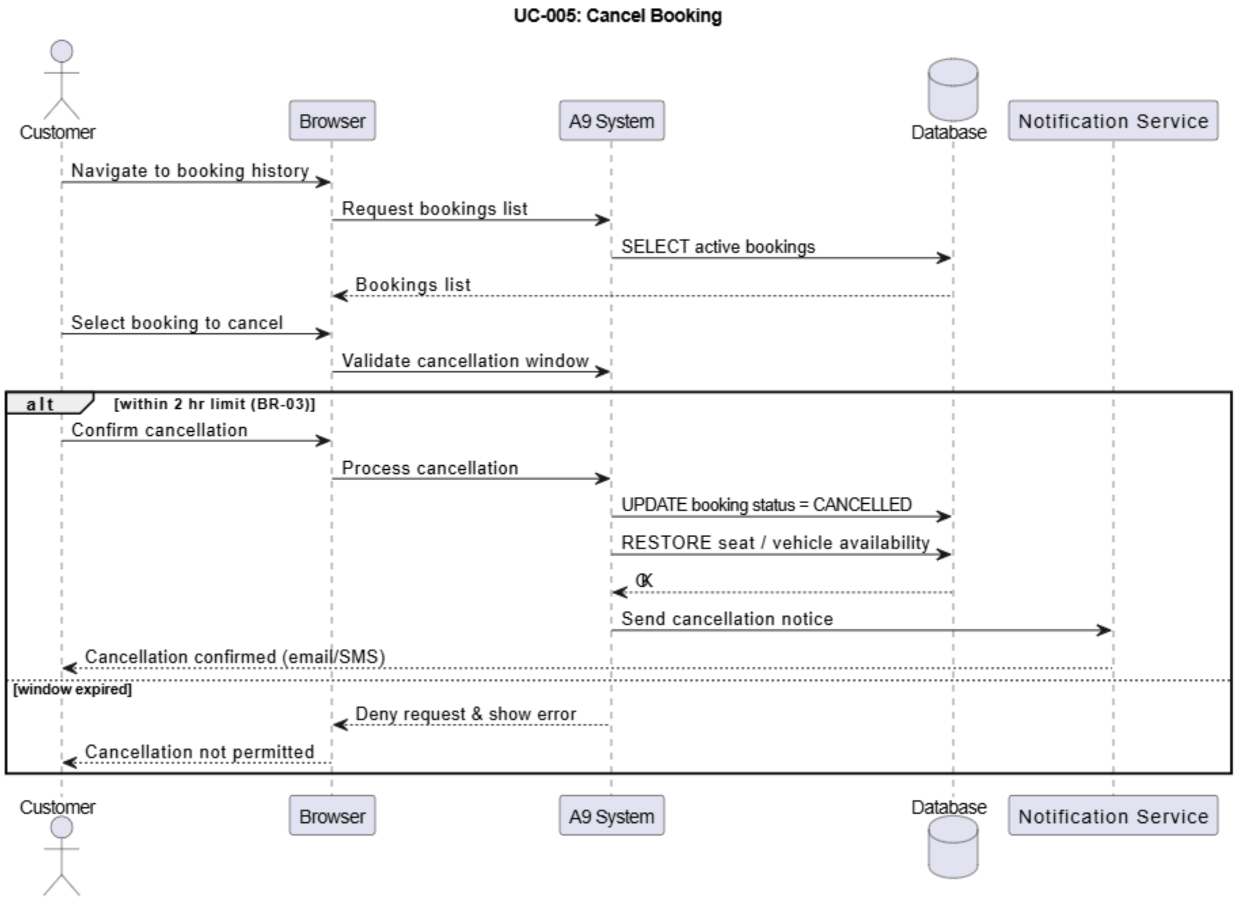
9.2.3 Admin Route & Vehicle Management



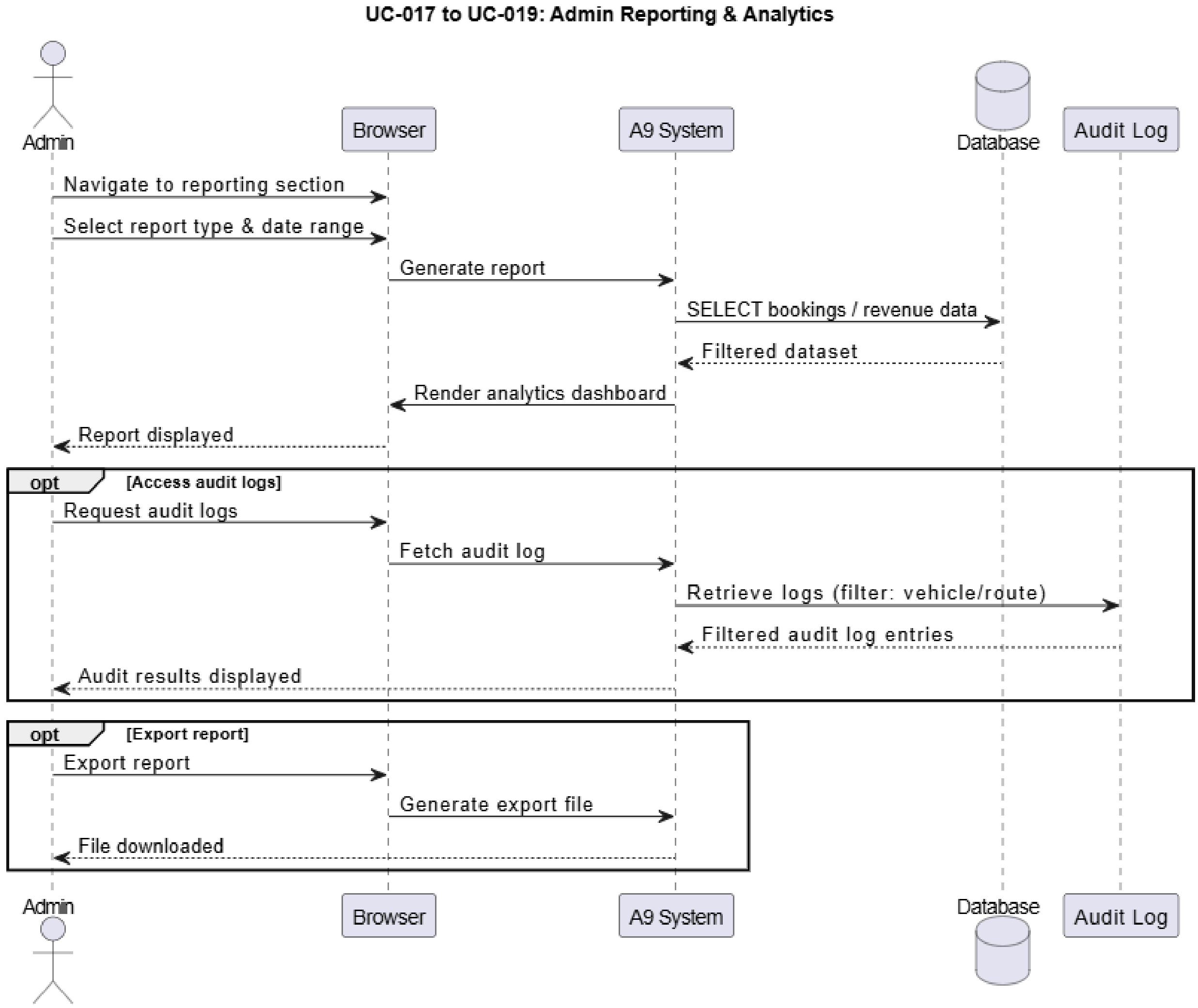
9.2.4 Notification & System Events



9.2.5 Cancel Booking



9.2.6 Admin Reporting & Analytics



10. CHAPTER
EXTERNAL INTERFACE REQUIREMENTS

10.1 User Interface Requirements

The A9 Express – Transport Booking Platform shall provide a responsive, user-friendly, and accessible interface for customers, transport operators, drivers, and administrators.

The interface shall:

- Be accessible through desktop and mobile devices
- Provide responsive layouts for different screen sizes
- Display real-time booking availability and updates
- Support simple navigation and efficient booking workflows
- Provide dashboards for customers, operators, and administrators

Customer Interface Requirements

Customers shall be able to:

- Register and log into the platform
- Search available transport services
- Book private bus tickets
- Reserve airport transfer services
- Create customized Sri Lankan tour packages
- Select tourist destinations and transportation options
- View booking confirmations and booking history
- Cancel or modify bookings based on service policies
- Receive notifications and travel updates

Operator and Administrator Interface Requirements

Operators and administrators shall be able to:

- Manage buses, routes, and schedules
- Configure pricing and availability
- Manage drivers and vehicles
- Monitor bookings and cancellations
- Access customer and operational reports
- Manage notifications and system settings

10.2 Hardware Interface Requirements

The system shall support integration with the following hardware components:

Hardware Component	Purpose
Smartphones	Customer, driver, and operator access
Desktop / Laptop Computers	Administrative and operational management
GPS-enabled Devices	Real-time location and route tracking
Cloud Servers	Application hosting and data storage
Internet Routers / Network Devices	Communication and network connectivity

10.3 Software Interface Requirements

The A9 Express platform shall integrate with various external software services and technologies required for system functionality.

Software Interface	Purpose
Payment Gateway API	Online payment processing
Google Maps API	Route calculation and location services
SMS Gateway	Sending booking and notification messages
Email Service	Sending booking confirmations and alerts
Database Management System	Data storage and retrieval
Authentication Service	User login and access management

Proposed Technology Stack

Component	Technology
Frontend	HTML, CSS, JavaScript, React
Backend	Node.js / Express
Database	MySQL / PostgreSQL
Mobile Application	React Native

10.4 Communication Interface Requirements

The platform shall support secure and reliable communication between users, servers, and integrated external services.

Communication Requirements

- HTTPS shall be used for secure web communication.
- SSL/TLS encryption shall be implemented for secure data transmission.
- REST APIs shall be used for communication between frontend and backend components.
- SMS and email services shall support customer notifications and confirmations.
- Real-time booking and availability updates shall be supported through internet-based communication.

Network Requirements

- Minimum 4G mobile data or broadband internet connectivity shall be required.
- The platform shall support stable communication under normal network conditions.
- Communication with third-party APIs shall use secure authenticated connections.

11. CHAPTER
DATA REQUIREMENTS

This chapter defines the data entities, attributes, relationships, storage requirements, and validation rules for the Transport Booking Management System.

11.1 Data Entities and Attributes

11.1.1 Customer

Stores personal and account information for registered customers.

Field	Data Type	Constraints	Description
customer_id	VARCHAR(10)	PRIMARY KEY, UNIQUE, NOT NULL	Unique customer identifier
first_name	VARCHAR(100)	NOT NULL	Customer first name
last_name	VARCHAR(100)	NOT NULL	Customer last name
email	VARCHAR(150)	UNIQUE, NOT NULL	Customer email address
phone_number	VARCHAR(15)	NOT NULL	Contact number
password_hash	VARCHAR(255)	NOT NULL	Encrypted password
address	TEXT	NULLABLE	Residential address
registered_date	DATETIME	DEFAULT NOW()	Registration date
account_status	ENUM('Active','Inactive','Blocked')	DEFAULT 'Active'	Customer account status

11.1.2 User (System Users)

Stores authentication and role information for Admins, Operators, Drivers, and Staff.

Field	**Data Type**	**Constraints**	**Description**
user_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique user ID
username	VARCHAR(50)	UNIQUE, NOT NULL	Login username
password_hash	VARCHAR(255)	NOT NULL	Encrypted password
role	ENUM('Admin','Operator','Driver','Staff')	NOT NULL	User role
email	VARCHAR(150)	UNIQUE, NOT NULL	User email
is_active	BOOLEAN	DEFAULT TRUE	Account status
failed_attempts	INT	DEFAULT 0	Failed login attempts
locked_until	DATETIME	NULLABLE	Account lock expiry
last_login	DATETIME	NULLABLE	Last login timestamp

11.1.3 Bus Route

Stores route information for bus transportation services.

Table 10-3: Bus Route Entity Specification

Field	Data Type	Constraints	Description
route_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique route ID
origin	VARCHAR(100)	NOT NULL	Starting location
destination	VARCHAR(100)	NOT NULL	Destination location
stops	TEXT	NULLABLE	Intermediate stops
distance_km	DECIMAL(6,2)	NULLABLE	Route distance
status	ENUM('Active','Inactive')	DEFAULT 'Active'	Route availability

11.1.4 Schedule

Stores bus schedule and trip details.

Table 10-4: Schedule Entity Specification

Field	Data Type	Constraints	Description
schedule_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique schedule ID
route_id	INT	FOREIGN KEY → Bus Route	Assigned route
departure_time	DATETIME	NOT NULL	Departure date and time
arrival_time	DATETIME	NOT NULL	Expected arrival time
seat_capacity	INT	NOT NULL	Total seat capacity

11.1.5 Booking

Stores all customer booking information.

Table 10-5: Booking Entity Specification

Field	**Data Type**	**Constraints**	**Description**
booking_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique booking ID
customer_id	VARCHAR(10)	FOREIGN KEY → Customer	Customer making booking
schedule_id	INT	FOREIGN KEY → Schedule	Selected trip
booking_date	DATETIME	DEFAULT NOW()	Booking timestamp
total_amount	DECIMAL(10,2)	NOT NULL	Total booking amount
booking_status	ENUM('Confirmed','Cancelled','Pending','Completed')	DEFAULT 'Pending'	Booking status
payment_status	ENUM('Paid','Unpaid','Refunded')	DEFAULT 'Unpaid'	Payment status

11.1.6 Seat

Stores seat allocation details for bookings.

Table 10-6: Seat Entity Specification

Field	Data Type	Constraints	Description
seat_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique seat ID
schedule_id	INT	FOREIGN KEY → Schedule	Related schedule
seat_number	VARCHAR(10)	NOT NULL	Seat number
booking_id	INT	FOREIGN KEY → Booking, NULLABLE	Assigned booking
seat_status	ENUM('Available','Reserved','Booked')	DEFAULT 'Available'	Seat availability

11.1.7 Vehicle

Stores information about buses and vehicles.

Table 10-7: Vehicle Entity Specification

Field	Data Type	Constraints	Description
vehicle_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique vehicle ID

Field	Data Type	Constraints	Description
registration_number	VARCHAR(20)	UNIQUE, NOT NULL	Vehicle registration number
vehicle_type	VARCHAR(50)	NOT NULL	Bus/Van/Car type
capacity	INT	NOT NULL	Passenger capacity
status	ENUM('Available','Maintenance','Assigned')	DEFAULT 'Available'	Vehicle status

11.1.8 Driver

Stores driver information.

Table 10-8: Driver Entity Specification

Field	Data Type	Constraints	Description
driver_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique driver ID
first_name	VARCHAR(100)	NOT NULL	Driver first name
last_name	VARCHAR(100)	NOT NULL	Driver last name
license_number	VARCHAR(50)	UNIQUE, NOT NULL	Driving license number

Field	Data Type	Constraints	Description
contact_number	VARCHAR(15)	NOT NULL	Driver contact number
status	ENUM('Available','Assigned','Inactive')	DEFAULT 'Available'	Driver availability

11.1.9 Tour Package

Stores customized tour package information.

Table 10-9: Tour Package Entity Specification

Field	Data Type	Constraints	Description
package_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique package ID
customer_id	VARCHAR(10)	FOREIGN KEY → Customer	Customer creating package
destinations	TEXT	NOT NULL	Selected destinations
duration_days	INT	NOT NULL	Tour duration
vehicle_type	VARCHAR(50)	NOT NULL	Selected vehicle
total_price	DECIMAL(10,2)	NOT NULL	Package price
package_status	ENUM('Pending','Confirmed','Cancelled')	DEFAULT 'Pending'	Package status

11.1.10 Audit Log

Records system activities for security and monitoring purposes.

Table 10-10: Audit Log Entity Specification

Field	Data Type	Constraints	Description
log_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique log ID
user_id	INT	FOREIGN KEY → User	User performing action
action_type	VARCHAR(100)	NOT NULL	Type of action
entity_type	VARCHAR(100)	NULLABLE	Affected entity
entity_id	VARCHAR(50)	NULLABLE	Related record ID
description	TEXT	NULLABLE	Action description
timestamp	DATETIME	DEFAULT NOW()	Action timestamp

11.2 Data Relationships

The key relationships between data entities are as follows:

- A Customer can have many Bookings (one-to-many).
- A Bus Route can have many Schedules (one-to-many).
- A Schedule can have many Bookings (one-to-many).
- A Booking can contain multiple Seats (one-to-many).
- A Vehicle can be assigned to many Schedules (one-to-many).
- A Driver can be assigned to many Schedules or Tour Packages (one-to-many).
- A Customer can create multiple Tour Packages (one-to-many).
- A User can perform many system actions recorded in the Audit Log (one-to-many).
- A Schedule references one Bus Route.

11.3 Data Storage and Retention

- All data shall be stored in a MySQL 8.0 relational database.
- Customer and booking records shall be retained for a minimum of 5 years.
- Audit logs shall be retained for a minimum of 90 days for security monitoring.
- Automated database backups shall be performed every 24 hours.
- Cancelled booking records shall not be permanently deleted to maintain historical tracking.
- Payment and booking transaction data shall be securely stored for auditing purposes.
- Sensitive information such as passwords shall be encrypted before storage.
- Tour package and airport transfer records shall be retained for reporting and analytics purposes.

11.4 Data Integrity and Validation Rules

- Customer IDs and booking IDs must be unique and system-generated.
- Email addresses must be unique for each registered customer and user.
- Seat bookings must prevent double booking of the same seat.
- Booking dates and departure times must not be set in the past.
- Available seat counts must never become negative.
- All monetary values shall use DECIMAL(10,2) data types.
- Foreign key constraints shall be enforced to maintain relational integrity.
- Passwords must be securely hashed before storage.
- Route, booking, and payment updates shall create corresponding Audit Log entries.
- Only authenticated users with valid roles shall access protected system modules.

12. CHAPTER
ALTERNATIVE SOLUTIONS CONSIDERED

Alternative	Description	Reason for Rejection
Manual / Paper-Based Booking System	Continue managing bus bookings, schedules, customer details, and tour reservations using paper records and registers.	Prone to human errors; difficult to track bookings and schedules; no real-time seat availability; inefficient data management; time-consuming.
Phone-Based Booking System	Customers make bookings through phone calls without using an online booking platform.	High chance of booking errors and double bookings; difficult to manage large booking volumes; no automated confirmations or notifications.
Spreadsheet-Based Management System	Use spreadsheet software to manage routes, bookings, vehicles, and customer records.	Not suitable for multi-user environments; lacks real-time updates; difficult to maintain data consistency; no automation features.
Separate Systems for Bus Booking, Airport Transfers, and Tours	Use independent systems for different transport services instead of a unified platform.	Lack of integration; duplicate data management; inefficient operations; difficult reporting and administration.
Mobile Application Only	Develop only a native mobile application without a web-based system.	Higher development and maintenance costs; requires separate applications for Android and iOS; limited accessibility for some users and administrators.
Admin Approval for Every Booking	Require admin approval before confirming every customer booking request.	Causes delays in booking confirmation; increases administrative workload; reduces

Alternative	Description	Reason for Rejection
		customer convenience and booking efficiency.
Offline Booking Management System	Manage bookings and schedules without internet connectivity.	No real-time synchronization; difficult to update seat availability and notifications; limited accessibility for customers and operators.
Cash-Only Payment Method	Allow customers to make payments only through cash transactions.	Reduces convenience for online users; limits digital booking capabilities; not suitable for customers preferring online payment methods.
Manual Tour Package Planning	Staff manually prepare customized tour packages for each customer request.	Time-consuming; difficult to calculate pricing accurately; inefficient for handling multiple customer requests simultaneously.

13. CHAPTER
FEASIBILITY STUDY

This chapter assesses the feasibility of the Transport Booking Management System across four dimensions: technical, economic, operational, and schedule feasibility.

13.1 Technical Feasibility

Technical feasibility evaluates whether the proposed system can be developed using available technologies, tools, and the development team's existing technical knowledge.

13.1.1 Technology Stack Assessment

Table 12-1: Proposed Technology Stack and Justification

Component	Proposed Technology	Justification
Frontend	React.js	Component-based architecture, responsive UI, and strong community support
Backend	Node.js	Scalable, secure, and suitable for REST API development
Database	MySQL 8.0	Reliable relational database suitable for booking and transaction management
Authentication	JWT + BCrypt	Secure authentication and password encryption
Containerization	Docker	Consistent deployment and environment management
Notifications	Email API / SMS API	Supports booking confirmations and alerts
API Documentation	Swagger / OpenAPI	Standardized and auto-generated API documentation

13.1.2 Technical Feasibility Assessment

- The development team has prior experience with React.js, Npde.js, and MySQL through academic projects and coursework.
- All selected technologies and frameworks are open-source and widely documented.
- The system architecture separates frontend, backend, and database layers for easier maintenance and scalability.
- Docker can be used to ensure consistent deployment environments across development and production systems.
- Real-time seat booking and availability management introduce moderate technical complexity but are achievable using database synchronization and API validation techniques.
- Integration with email or SMS notification services can be implemented using existing third-party APIs.

Conclusion: The project is technically feasible. The selected technologies are modern, reliable, and within the technical capabilities of the development team.

13.2 Economic Feasibility

Economic feasibility evaluates whether the system can be developed and maintained within acceptable financial limits compared to its expected benefits.

13.2.1 Cost Estimate

Table 12-2: Cost Estimation

Cost Item	Estimated Cost (LKR)
Development Cost (student project – no labour cost)	0
Cloud Hosting (12 months)	~60,000 – 120,000
Domain Name Registration (1 year)	~3,000 – 5,000
SSL Certificate (Let's Encrypt – free)	0
SMS / Email Notification Service	~10,000 – 20,000 per year

Cost Item	Estimated Cost (LKR)
Software Licenses (Open-source stack)	0
Total Estimated Annual Operational Cost	~73,000 – 145,000 LKR

13.3 Benefit Analysis

- The system reduces manual booking and scheduling operations.
- Real-time booking management minimizes booking conflicts and double bookings.
- Automated notifications improve customer communication and service reliability.
- Digital records improve route, vehicle, and customer management efficiency.
- Online booking capabilities improve customer convenience and business accessibility.
- Reporting and analytics features help operators make better business decisions.

Conclusion: The system is economically feasible. The operational costs are reasonable compared to the expected improvements in efficiency, customer satisfaction, and business management.

13.4 Operational Feasibility

Operational feasibility evaluates whether the system can be effectively used within the organization's daily operations.

User Readiness

- Customers and staff are assumed to possess basic computer and smartphone literacy.
- The system interface is designed to be simple and user-friendly.
- Key operations such as booking tickets, managing schedules, and viewing reports require minimal training.

13.4.1 Organizational Impact

- Existing manual booking processes will be replaced with digital workflows.
- Initial staff training sessions will be required before system deployment.
- The system does not require significant hardware changes beyond existing computers and internet access.
- Admin users can manage schedules, pricing, vehicles, and user accounts without advanced technical skills.

Conclusion: The system is operationally feasible. The interface design and automation features support smooth adoption by customers and transport operators.

13.5 Schedule Feasibility

Schedule feasibility evaluates whether the project can be completed within the planned timeline.

13.5.1 Estimated Project Timeline

Table 12-3: Estimated Project Timeline

Phase	Description	Duration	Milestone
1	Requirements Analysis & SRS Documentation	3 Weeks	SRS Approved
2	System Design (Architecture, Database, UI Mockups)	3 Weeks	Design Completed
3	Frontend Development (React.js Modules)	5 Weeks	UI Prototype
4	Backend Development (APIs, Authentication, Database Integration)	5 Weeks	APIs Tested
5	Integration Testing & Bug Fixing	3 Weeks	Test Reports
6	User Acceptance Testing (UAT)	2 Weeks	UAT Sign-off
7	Deployment & Final Documentation	2 Weeks	System Deployment
8	Buffer Time / Contingency	2 Weeks	Project Completion

Total Estimated Duration: 25 Weeks

13.5.2 Schedule Risk Factors

- A multi-member development team allows parallel frontend and backend development.
- Academic workload and examinations may temporarily affect development progress.
- Modular system architecture supports phased implementation and testing.
- Clearly defined project scope helps reduce scope creep and scheduling delays.

Conclusion: The schedule is feasible. The estimated timeline is achievable provided that tasks are properly managed and project scope is controlled effectively.

14. CHAPTER
RISK ANALYSIS

This section identifies and evaluates potential risks associated with the development of the Transport Booking Management System. The analysis focuses on technical, resource, schedule, and operational risks that may impact project execution and system delivery. Appropriate mitigation strategies have been defined to minimize the likelihood and impact of each identified risk.

Table 14-1: Risk Analysis

Risk ID	Description	Category	Likelihood	Impact	Risk Level	Mitigation Strategy
R1	Integration issues between frontend and backend components	Technical	Medium	High	High	APIs shall be defined early, and integration testing shall be performed continuously during development.
R2	Delays due to academic workload and internship commitments	Schedule	High	Medium	High	A realistic project schedule shall be maintained, and tasks shall be prioritized properly.
R3	High occurrence of software defects during development	Technical	High	Medium	High	Code reviews, testing, and debugging practices shall be carried out

Risk ID	Description	Category	Likelihood	Impact	Risk Level	Mitigation Strategy
						throughout development.
R4	Real-time seat availability failures causing double bookings	Operational	Medium	High	High	Booking validation and real-time synchronization mechanisms shall be implemented.
R5	Security vulnerabilities affecting customer and booking data	Technical	Medium	High	High	Authentication, authorization, encrypted passwords, and secure APIs shall be implemented.
R6	Failure of third-party services such as SMS, email, or payment gateways	External	Medium	Medium	Medium	Reliable third-party providers and fallback notification methods shall be considered.

15. CHAPTER

References

CLIENT COLLABORATION AGREEMENT

Date: 22nd April 2026

To Whom It May Concern,

This letter confirms that **Amirthalingam Nirmalan**, representing **Annai Muththumari Express**, has agreed to collaborate with the following students from the Faculty of Science, University of Kelaniya, Sri Lanka:

- Dayastan Thevarasa (SE/2022/007)
- Arushanth Vasanthakumar (SE/2022/016)
- Ronald Ensilirukman (SE/2022/042)
- Sathurjan Surenthiran (SE/2022/035)
- Arulanantham Mathumithan (SE/2022/015)

This collaboration is undertaken as part of their academic **System Design Project.**

The purpose of this collaboration is to design and develop an **Online Transport Booking Platform** that will support private bus ticket reservations, airport transfers, and tour package bookings, while enhancing operational efficiency and improving customer convenience.

1.Project Title

A9 Express - Online Transport Booking Platform

2. Background and Problem Statement

At present, transport booking services such as private bus reservations, airport transfers, and tour package bookings are handled through disconnected systems or manual processes. Existing platforms offer limited flexibility for transport operators, including restricted branding opportunities, limited pricing control, and inadequate support for value added services.

As a result, customers face inconvenience during booking, while transport providers struggle to manage services efficiently and build direct customer relationships.

To address these issues, there is a need for a centralized online transport booking platform that enables efficient booking management, flexible service offerings, and improved user experience for both customers and operators.

3. Objectives of the Proposed System

The proposed system aims to provide a user-friendly and efficient online platform for customers to book private bus tickets with ease. It will also enable airport transfer bookings with flexible scheduling options to suit different travel needs. In addition, the system allows customers to conveniently browse, select, and reserve tour packages based on their preferences. For operators, the platform offers efficient management of routes, bookings, pricing, and schedules in one centralized system. Overall, the proposed solution seeks to enhance customer satisfaction by offering real-time updates, secure online booking, and a smooth travel reservation experience.

4. Scope of Collaboration

The client agrees to provide relevant information regarding current booking and operational procedures, and to actively participate in requirement gathering and clarification meetings. They will also review and validate system requirements and design documents and provide constructive feedback on both the prototype and the final solution to ensure the system meets their needs.

The student team agrees to use all client provided information strictly for academic purposes and to maintain strict confidentiality of any sensitive data. They will provide regular updates on project progress and ensure the client is informed of any significant changes to the agreed project scope throughout the development process.

5. Nature of the Project

This project is carried out strictly for academic purposes as part of the students' university coursework. The client acknowledges that the software being developed is a student academic project intended solely for learning and demonstration purposes. It is understood that the **University of Kelaniya** and the student team bear no responsibility for any direct or indirect loss that may arise from the use of the developed system.

Furthermore, the client agrees that no commercial warranties or guarantees are provided for this academic delivery. The system is offered **"as is"** for educational evaluation, and any use beyond the academic scope is at the client's own discretion and risk.

6. Confidentiality

Any confidential information shared by the client during the project shall be used strictly for academic project purposes and handled with appropriate care. The student team agrees to protect such information from any unauthorized disclosure and ensure it is securely managed throughout the project lifecycle.

In addition, no confidential information will be shared with any third parties without prior approval from the client. These confidentiality obligations will remain valid for a **period of two (2) years** after the completion of the project.

7. Intellectual Property Rights

All software, documentation, system designs, and project deliverables developed under this project shall remain the intellectual property of the **student developers and/or the University of Kelaniya**.

The client may use the developed system **only for review and academic evaluation purposes**, unless otherwise agreed in writing.

8. Agreement Duration

This agreement becomes effective on the signing date and remains valid until the academic project is completed and submitted.

Either party may terminate the agreement by providing **14 days' written notice.**

9. Confirmation

I hereby confirm that the above project request is genuine and that the proposed solution is required as described.

Client Details

Client Name: Amirthalingam Nirmalan

Designation: Owner

Organization: Nirmalan A9 Express

Contact Number / Email: 077 6311920

Signature: A Nirme

Official Seal: **Nirmalan A9 Express**
Travels & Tour
Reg.No: KVD/1755
0776311920

Date: 23/04/2026

PROJECT APPROVAL FORM

SENG 31242 – System Design Project

IMPORTANT: This form must be signed by the Design Project Supervisor following the Idea Pitch. A signed copy must be committed to the team's documents/pitch/ repository before any further project work commences.

SECTION 1 — PROJECT DETAILS

Team Name	Cloud Creaters
Project Name	A9 Express – Transport Booking Platform
Project Brief	A9 Express is a unified online platform for private bus bookings, airport transfers, and customizable Sri Lankan tour package reservations.
Academic Year	2024/25
Project Type	Real Client
Client / OS Project	Name :A9 Express [Veerapaththirar Kovilady, Uduppiddy, Jaffna]
Team Members	Thevarasa Dayastan -SE/2022/007 Arulanantham Mathumithan -SE/2022/015 Vasanthakumar Arushanth -SE/2022/016 Surenthiran Sathurjan -SE/2022/035 Ensilirukman Ronald -SE/2022/043

SECTION 2 — PANEL DECISION

☑ **APPROVED** *Team is authorised to proceed to Project Kick-Off.*	☐ **NOT APPROVED** *Re-pitch must be scheduled within 3 working days.*

Conditions (if any) –

Panel Feedback –

SECTION 3 — AUTHORISATION SIGNATURE

Design Project Supervisor
Eng. Dr. Tiroshan Madushanka
Software Engineering Teaching Unit, Faculty of Science, University of Kelaniya

Signature:

Date: 21/05/2026 .

Software Engineering Teaching Unit
Faculty of Science
University of Kelaniya
Kelaniya Sri Lanka